

Unit 5 - Triangle Congruence	
Lesson	Page Number
5.1 Exploring Side-Side-Side Triangle Congruence	223
5.2 Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence	229
5.3 Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence	234
5.4 Exploring Corresponding Parts of Congruent Triangles are Congruent	241
5.5 Drawing Conclusions from Given Information	246
5.6 Proving Triangles Congruent	252
5.H1 Definition of Congruence in Terms of Rigid Transformations	257
5.7 Corresponding Parts of Congruent Triangles are Congruent	262
5.8 The Proof is in the Bag!	267
5.9 Congruence in Two-Dimensional Figures – Part 1	272
5.10 Congruence in Two-Dimensional Figures – Part 2	276

The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

- *Clarification 1: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.*
- *Clarification 2: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

- *Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.*

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- *Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.*
- *Clarification 2: Transformations include translations, dilations, rotations, and reflections described using words or using coordinates.*
- *Clarification 3: Within the Geometry course, rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation.*
- *Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.*

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

- *Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.*

MA.912.GR.5.1 Construct a copy of a segment or an angle.

- *Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.*

MA.912.GR.2.7 (Honors only) Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

Unit 5, Lesson 1: Exploring Side-Side-Side Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Target

- ☐ I can use constructions and transformations to determine if SSS congruence can be used to show that two triangles are congruent.

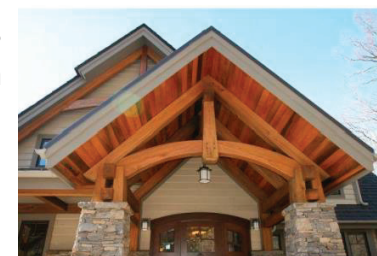


5.1.1 Warm-Up: STEM

Triangles are effective tools for architecture used in the design of buildings and other structures, as they provide strength and stability.

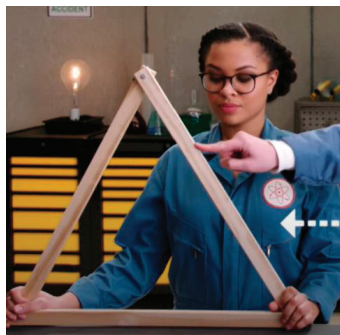


When building materials are used to form a triangle, the design has a heavy base and the pinnacle on the top is capable of handling weight because of how the energy is distributed throughout the triangle. The triangle's use in architecture dates back longer than other common architectural shapes such as the dome, arch, and cylinder. It even predates the wheel.



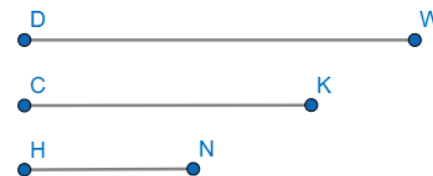
1. Imagine using straws or toothpicks to make a square or rectangle. What do you think will happen when the engineer puts pressure on the side of the rectangle?

2. Now imagine you made a triangle using straws or toothpicks. What do you think will happen when the engineer puts pressure on the side of the triangle?



5.1.2 Exploration: Are Three Sides Enough?

1. $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ are given. Use the given steps to construct $\triangle BYT$ so that its sides are congruent to $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$.



Step 1: Copy $\overline{DW}$. Construct $\overline{BY}$ so that it is congruent to $\overline{DW}$.

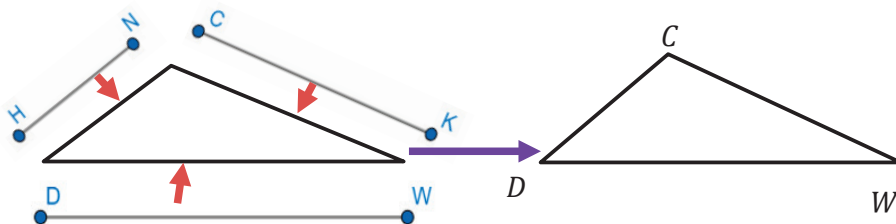
Step 2: Copy $\overline{CK}$. Place the compass on point B to create an arc that is the length of $\overline{CK}$.

Step 3: Copy $\overline{HN}$. Place the compass on point Y to create an arc that is the length of $\overline{HN}$.

Step 4: Mark the intersection of the two arcs using a point. Label the point T .

Step 5: Draw $\overline{BT}$ and $\overline{YT}$.

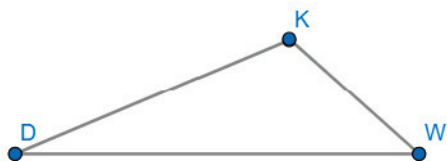
2. The triangle below was created using the three given line segments.



Compare the triangle you constructed with $\triangle CDW$. Describe any similarities or differences.

3. Maria used a computer program to create line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$. She then moved the line segments to construct $\triangle DKW$. Because the vertices of the line segments $\overline{CK}$ and $\overline{HN}$ coincided with other vertices, she only used the letters D , K , and W .

- a. What do you notice about $\triangle BYT$ and $\triangle DKW$?

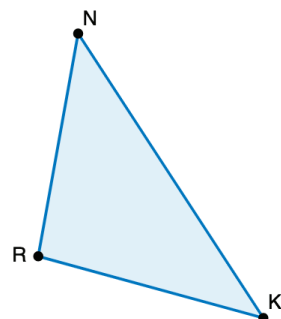
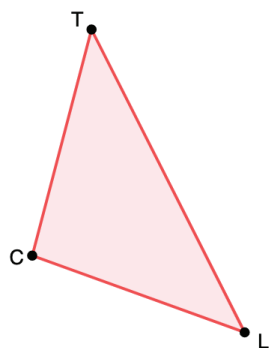


- b. The line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ were used to construct three different triangles. Work with your partner to write a conjecture about creating triangles whose sides are congruent.

- c. Ask two classmates about their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture for Creating Triangles Whose Sides are Congruent	My Summary of Their Reason	Initials

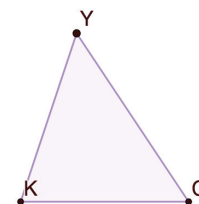
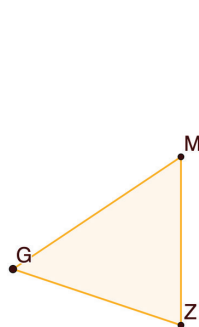
4. Use patty paper to determine if $\triangle CTL$ and $\triangle RNK$ were created using the same three line segments.



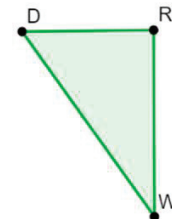
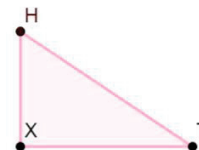
5. Discuss the following with your partner: If three sides of a triangle are congruent, does that mean all of the angles of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

6. Two sets of triangles are shown.

$\triangle MGZ$ and $\triangle KYC$

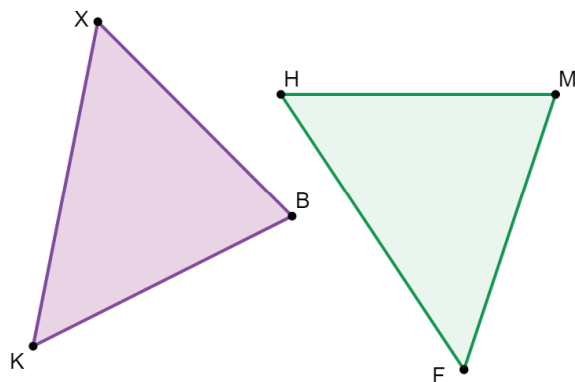


$\triangle HTX$ and $\triangle WDR$



- Use patty paper to determine if the sides of the two triangles are congruent.
- $\triangle$ _____ and $\triangle$ _____ are congruent because all of the sides are congruent.
- For the two triangles that are congruent, use patty paper to verify that the angles are congruent.
- For the two triangles that are congruent, list the sides on each triangle that are congruent to each other.

7. $\triangle KXB$ and $\triangle HFM$ are shown. $HM = 4.2$, $XB = 4.2$, $XK = 5$, and $KB = 4.4$.



- a. Discuss with your partner if the given measurements are sufficient to determine whether $\triangle KXB$ and $\triangle HFM$ are congruent. Explain your decision.
- b. Use patty paper, a protractor, a ruler, or a compass to measure all of the parts of $\triangle KXB$ and $\triangle HFM$. Then, complete the sentence.

$\triangle KXB$ and $\triangle HFM$ are

- ☐ congruent
☐ not congruent

because

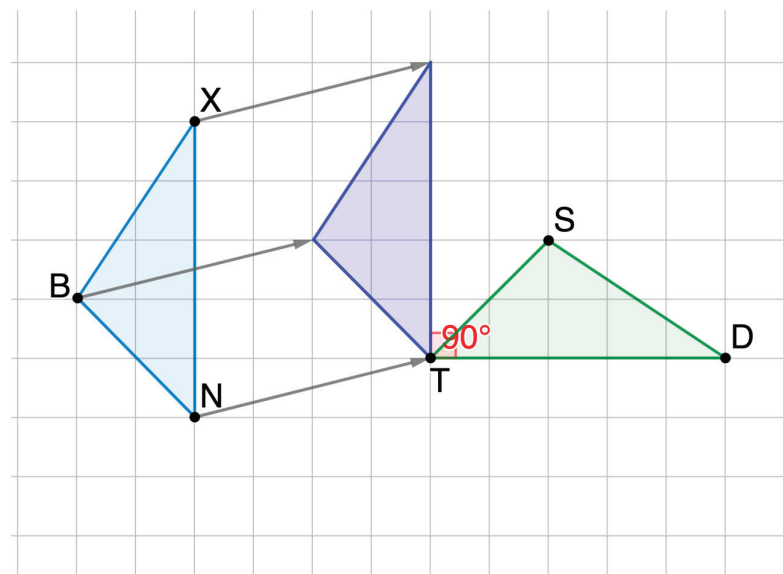
side(s) and

- ☐ 0
☐ 1
☐ 2
☐ 3

angle(s) are congruent.

- ☐ 0
☐ 1
☐ 2
☐ 3

8. A series of transformations shows how $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated, then rotated 90° about point T to create $\overline{SD}$, $\overline{DT}$, and $\overline{TS}$.



- a. Use the grid to determine if $BN = TS$, $BX = DS$, and $XN = TD$.
- b. Discuss with your partner how the transformations that were used ensure that the length of each line segment was preserved.

- c. Use the word bank to complete the statements. Words may be used more than once.

Word Bank				
angles	congruent	congruence	parts	rigid motions

Since $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated then rotated 90° , the length of each line segment was preserved because of the definition of congruence in terms of _____. The definition of congruence in terms of _____ also ensures that the _____ are congruent. Because all of the _____ of $\triangle XNB$ are _____ to all of the corresponding _____ of $\triangle DTS$, the two triangles are _____.



5.1.3 Bring It Together: Triangle Congruence



Side-Side-Side (SSS) Congruence Postulate

Two triangles are congruent if the three sides of one are congruent to the three sides of the other, respectively.

1. Work with your partner to explain how the exploration has shown that the Side-Side-Side (SSS) Congruence Postulate is true.



5.1.4 Wrap-Up: Cool Down

1. Maggie and Naomi both constructed a triangle using the side lengths shown.

- $CN = 2.5$ inches (in.)
- $NK = 3.75$ in.
- $CK = 5.25$ in.

Explain whether it is possible to find out if the two triangles Maggie and Naomi created will be congruent.



Unit 5, Lesson 2: Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



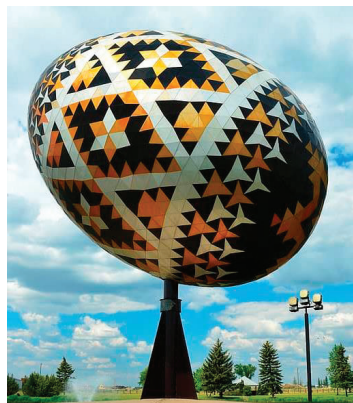
Learning Targets

- ☐ I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent.
- ☐ I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.



5.2.1 Warm-Up: Geometry in Art

A Pysanka is a decorated egg that is traditional in Ukraine. The decoration on the eggs is created using a wax resist (batik) method. Families often have their own unique pattern. The largest Pysanka is in Vegreville, Canada. It weighs 5,000 pounds, is 31 feet (ft) tall, and 18 ft wide. A traditional Pysanka egg is 2.5 inches tall and 1.75 inches wide.



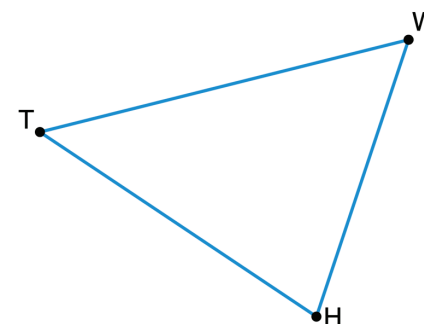
1. What do you notice about Pysanka eggs?
2. Estimate the number of traditional Pysanka eggs that will fit into the largest Pysanka. Show your work or explain how you found your estimate.



5.2.2 Exploration: Congruence Relationships

1. Triangle TWH is shown.

- a. Construct $\triangle XMP$ so that $\overline{XM} \cong \overline{TW}$, $\overline{PM} \cong \overline{HW}$, and $\angle TWH \cong \angle PMX$ using the given steps.



Construction	Steps
	1. Copy $\overline{TW}$ using a compass to construct $\overline{XM}$ using the orange line so that $\overline{XM} \cong \overline{TW}$.
	2. Copy $\angle TWH$ to construct $\angle PMX$ so that $\angle TWH \cong \angle PMX$.
	3. Copy $\overline{HW}$ to construct $\overline{PM}$ so that $\overline{PM} \cong \overline{HW}$.
	4. Draw $\overline{PX}$.

- b. Use patty paper to determine if $\triangle XMP$ is congruent to $\triangle TWH$.
- c. Discuss the following with your partner: If two sides of a triangle and their included angles are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

2. Construct $\triangle KSB$ so that $\overline{KS} \cong \overline{TH}$, $\overline{SB} \cong \overline{TW}$, and $\angle TWH \cong \angle SBK$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{KS}$ using the red line so that $\overline{TH} \cong \overline{KS}$.
	2. Copy $\overline{TW}$ to construct $\overline{SB}$ so that $\overline{TW} \cong \overline{SB}$.
	3. Copy $\angle TWH$ to construct $\angle SBK$ so that $\angle TWH \cong \angle SBK$.
	4. Draw $\overline{BK}$.

- Use patty paper to determine if $\triangle KSB$ is congruent to $\triangle HTW$.
- Discuss the following with your partner: If two sides of a triangle and a non-included angle are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

3. Construct $\triangle RYV$ so that $\overline{VY} \cong \overline{TH}$, $\angle WTH \cong \angle RYV$, and $\angle WHT \cong \angle RYV$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{VY}$ using the green line so that $\overline{TH} \cong \overline{VY}$.
	2. Copy $\angle WHT$ to construct $\angle RYV$ so that $\angle WHT \cong \angle RYV$.
	3. Copy $\angle WTH$ to construct $\angle RYV$ so that $\angle WTH \cong \angle RYV$.
	4. Label the intersection of the sides of the two angles point R .

- Use patty paper to determine if $\triangle RYV$ is congruent to $\triangle WHT$.
- Discuss the following with your partner: If two angles of a triangle and their included side are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.



5.2.3 Bring It Together: Triangle Congruence

1. Complete the statements.

- a. $\triangle XMP$ was created using ☐ 1 ☐ 2 ☐ 3 side(s) and ☐ 1 ☐ 2 ☐ 3 angle(s).

The construction showed that it is ☐ possible ☐ not possible

to assume that if the ☐ 1 ☐ 2 ☐ 3 side(s) of a triangle

and their ☐ adjacent angle(s) ☐ included angle(s) ☐ opposite angle(s) are congruent,

then the triangles are congruent.

- b. $\triangle KSB$ was created using ☐ 1 ☐ 2 ☐ 3 side(s) and ☐ 1 ☐ 2 ☐ 3 angle(s).

The construction showed that it is

☐ possible ☐ not possible to assume that if the ☐ 1 ☐ 2 ☐ 3

side(s) of a triangle and the ☐ included angle(s) ☐ non-included angle(s)

are congruent, then the triangles are congruent.

- c. $\triangle RYV$ was created using ☐ 1 ☐ 2 ☐ 3 side(s) and ☐ 1 ☐ 2 ☐ 3 angle(s).

The construction showed that it is ☐ possible ☐ not possible

to assume that if the ☐ 1 ☐ 2 ☐ 3 side(s) of a triangle

and its ☐ adjacent angle(s) ☐ included angle(s) ☐ opposite angle(s) are congruent,

then the triangles are congruent.



Side-Angle-Side (SAS) Congruence Postulate

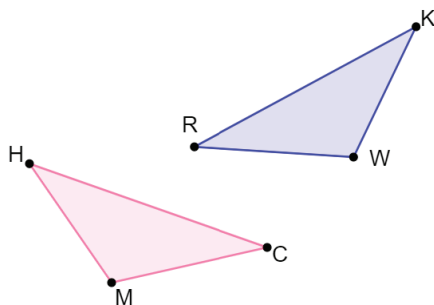
If two sides and the included angle of one triangle are congruent to two sides and the included angle of a second triangle, respectively, then the two triangles are congruent.



Angle-Side-Angle (ASA) Congruence Theorem

If two angles and an included side of one triangle are congruent respectively to two angles and an included side of a second triangle, then the two triangles are congruent.

2. Triangles HCM and RKW are shown. Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate to show that.

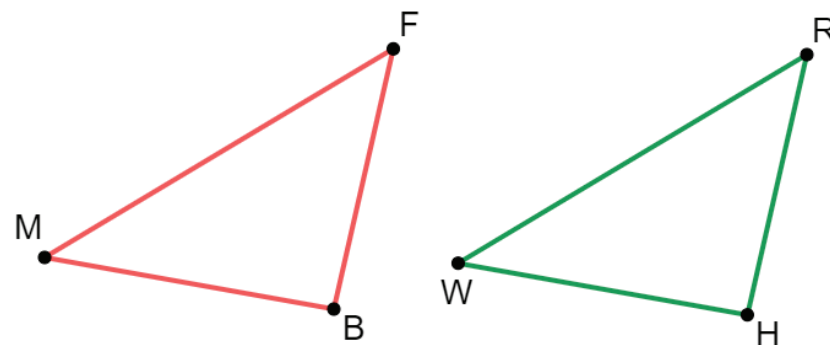


Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{MC} \cong \overline{RW}$, $\overline{HC} \cong \overline{KR}$, $\angle MCH \cong \angle WRK$		
$\overline{MC} \cong \overline{KW}$, $\overline{HM} \cong \overline{WR}$, $\overline{HC} \cong \overline{KR}$		
$\overline{KW} \cong \overline{HM}$, $\angle KRW \cong \angle CHM$, $\overline{MC} \cong \overline{RW}$		
$\overline{RK} \cong \overline{CH}$, $\angle MCH \cong \angle WRK$, $\angle MHC \cong \angle RKW$		
$\angle RKW \cong \angle CHM$, $\angle RWK \cong \angle CMH$, $\overline{KW} \cong \overline{HM}$		



5.2.4 Wrap-Up: Ticket Out the Door

Triangles BFM and WRH are shown.



1. Determine if the given information is sufficient to determine that the triangles are congruent. If so, state the congruence theorem or postulate to show that.

Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{FB} \cong \overline{HW}$, $\overline{FM} \cong \overline{RW}$, $\angle BFM \cong \angle HWR$		
$\overline{FB} \cong \overline{RH}$, $\angle FBM \cong \angle RHW$, $\overline{MB} \cong \overline{WH}$		
$\overline{FB} \cong \overline{HW}$, $\overline{BM} \cong \overline{HR}$, $\overline{FM} \cong \overline{RW}$		

Unit 5, Lesson 3: Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Targets

- ☐ I can use constructions to determine if AAS congruence can be used to show that two triangles are congruent.
- ☐ I can determine if HL congruence can be used to show that two triangles are congruent using deduction.



5.3.1 Warm-Up: Giving Back

Morgan State University is a historically black, public research university in Baltimore, Maryland. In the early 1960s, Calvin E. Tyler Jr., a young African American man who was the first in his family to go to college, attended Morgan State. However, in 1963 he was forced to drop out when he was unable to pay his tuition.



He began working as a truck driver for the United Parcel Service (UPS). Throughout his career at UPS, he continued to be promoted and ultimately served as both the senior vice president for U.S. operations and as a director within the company. On February 23, 2021, he and his wife Tina pledged \$20 million to Morgan State University's endowment to go toward scholarships for students in financial need.

Choose one of the following questions to answer.

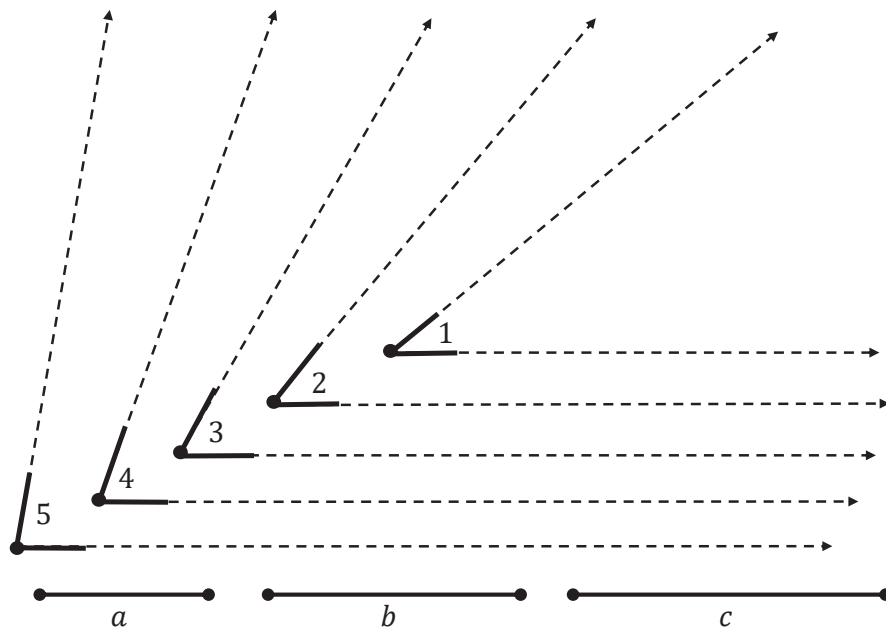
1. How do you think having a strong drive to succeed likely played a role in Mr. Tyler's career?
2. Why do you think it was important to Mr. Tyler to pledge this endowment?





5.3.2 Exploration: Which Other Criteria is Enough?

Five angles labeled 1-5 and three line segments labeled a , b , and c are given.



Use the given parts to create triangles by copying the parts needed onto patty paper using the given criteria.

Helpful Hint: When copying an angle, include some (or all) of the dotted lines so that angles and line segments can be connected.

1. Construct two different triangles using each set of criteria. Use the provided steps in each criteria.

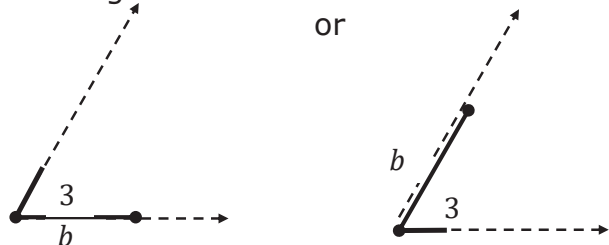
Criteria 1: Given two angles (AA)

- Step 1: Copy $\angle 1$ onto a piece of patty paper.
Step 2: Close the triangle using $\angle 2$.
Step 3: Trace the completed triangle. Label $\angle 1$ and $\angle 2$ in the completed triangle.
Step 4: Label your triangle "Criteria 1 (AA)."
Step 5: Repeat these steps with $\angle 3$ and $\angle 5$.

Criteria 2:**Given two angles and a non-included side (AAS)**

Step 1: Copy side b onto a piece of patty paper. Label side b on your patty paper.

Step 2: Place $\angle 3$ at one of the ends of side b and trace the ray that does not overlap with side b to copy the angle. Label $\angle 3$.



Step 3: Use $\angle 4$ to close the triangle such that $\angle 4$ is opposite from side b . Label $\angle 4$.

Step 4: Trace the completed triangle. Label your triangle "Criteria 2 (AAS)."

Step 5: Repeat these steps with side c , $\angle 1$, and $\angle 2$.

2. Compare your four triangles with your partner's.

3. Write your partner's name on the line in the table. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to _____'s triangle.				

4. With your partner, find another pair of students to form a group of four. Write the two new students' names in the table below. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to _____'s triangle.				
My triangle is congruent to _____'s triangle.				

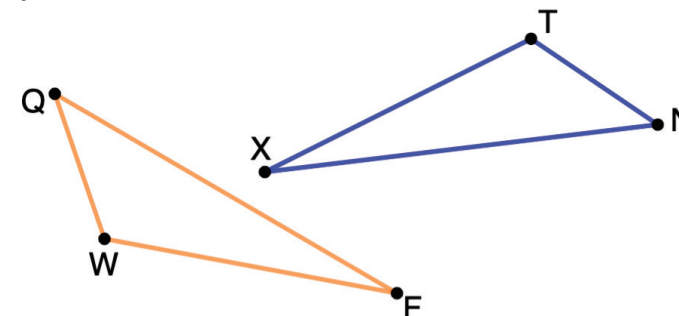
5. Discuss with your group: If given either set of criteria, does that mean all of the parts of the triangles are congruent? Then, write your explanation in the table.

Criteria	Explanation
Given two angles (AA)	
Given two angles and non-included side (AAS)	



5.3.3 Collaboration: Making Connections

1. $\triangle QWF$ and $\triangle NTX$ are shown. $\overline{FQ} \cong \overline{XN}$, $\angle FQW \cong \angle XNT$, and $\angle QWF \cong \angle NTX$.



Two students discussed the information given.

Mabel thinks there is not enough information to determine congruence, because the side given is not between the two angles given.

Xavier thinks the triangles are congruent because he knows that the third angles in the triangles have to be congruent, and then there would be a side between two angles.

With your partner, discuss both students' reasoning. Then, answer the following.

- a. Explain whether you agree with Mabel or Xavier.

- b. What additional information could be added to the argument you agree with to make it more convincing?

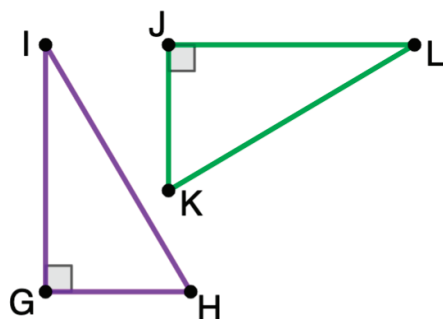
2. Triangle IHG and ΔLKJ are shown. $\overline{IH} \cong \overline{LK}$, $\overline{HG} \cong \overline{KJ}$, and $\angle IGH$ and $\angle LJK$ are right angles.

Eli and Kaylah had the following discussion.

Eli said, "In a situation where you have two sides without the angle in between them, you can't tell if the triangles are congruent."

Kaylah said, "Normally I would agree with you, except these are right triangles."

- a. Discuss with your partner what Kaylah might be thinking.
- b. When given two sides of a right triangle, how can the length of the third side be determined?



Kaylah continued, "For a right triangle, as long as I have two of the sides, I can find the length of the third side. Then I can use SSS since all three corresponding sides would be congruent."

- c. Work with your partner to explain why Kaylah's argument could work to show congruence for any pair of right triangles where the hypotenuse and one leg are identified as congruent.

- d. Explain whether Kaylah's argument would be needed to show congruence between right triangles where both legs of the triangles are identified as congruent.



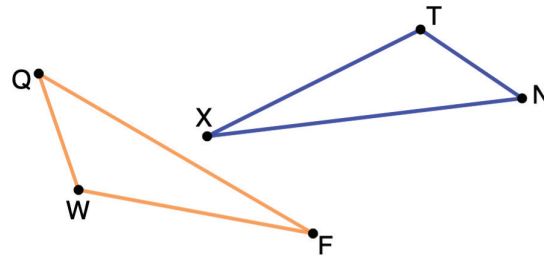
5.3.4 Bring It Together: Triangle Congruence



Angle-Angle-Side (AAS) Congruence Theorem

If two angles and a side opposite one of them are congruent to two angles and the corresponding side of the other, respectively, then the triangles are congruent.

1. Work with your partner to explain how Angle-Angle-Side Congruence Theorem is true. Use the triangles shown to provide an example if desired.



2. Complete the statements.

If two angles of one triangle are congruent to two

angles of another triangle, then the third pair of angles

are ☐ similar.
☐ congruent. If a congruent side opposite one of

the given angles is congruent to the corresponding side

of the other triangle, then ☐ SSS
☐ SAS
☐ ASA also applies. Thus,

AAS is enough to show congruence.



The Hypotenuse-Leg (HL) Theorem

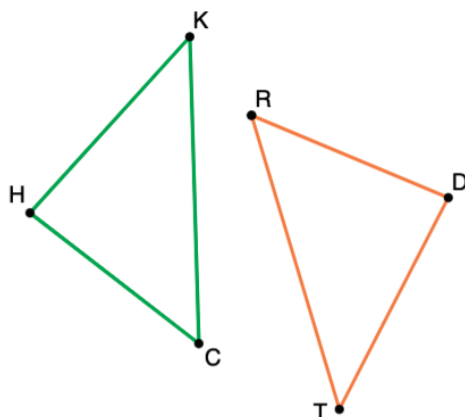
Two right triangles are congruent if the hypotenuse and a leg of one are congruent to the hypotenuse and a leg of the other, respectively.

3. The previous lesson's exploration showed that SSA is not sufficient to prove congruence. Explain why HL is sufficient to prove congruence for right triangles, despite being two sides with a non-included angle.



5.3.5 Try It: Which Criteria?

1. Triangles HKC and DTR are shown. Determine if the given information is sufficient to determine the triangles are congruent. If so, name the congruence theorem or postulate that shows that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\angle KCH \cong \angle TRD$		
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\angle CHK \cong \angle RDT$		
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\overline{HC} \cong \overline{DR}$		
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\overline{HK} \perp \overline{HC}$, $\overline{DT} \perp \overline{DR}$		
$\overline{HK} \cong \overline{DT}$, $\overline{HC} \cong \overline{DR}$, $\angle KHC \cong \angle TDR$		



5.3.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each congruence postulate or theorem.



I need help.



I got it.



I could help someone.

Side-Side-Side (SSS)			
Side-Angle-Side (SAS)			
Angle-Side-Angle (ASA)			
Angle-Angle-Side (AAS)			
Hypotenuse-Leg (HL)			

Unit 5, Lesson 4: Exploring Corresponding Parts of Congruent Triangles are Congruent



Benchmarks

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



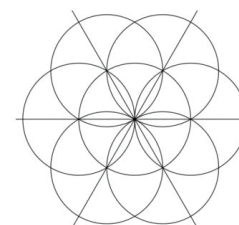
Learning Targets

- ☐ I can relate the definition of congruence in terms of rigid motions to corresponding parts of congruent triangles.
- ☐ I can use corresponding parts of congruent triangles to write congruence statements.



5.4.1 Warm-Up: Art in Geometry

Salih is taking an art class that looks at the different patterns in tiles from across the world. Salih used a compass to create this first step in a 12-fold pattern.



1. What do you notice about the design?

The last step of the 12-fold pattern and the final design are shown.

Last Step	Final Design

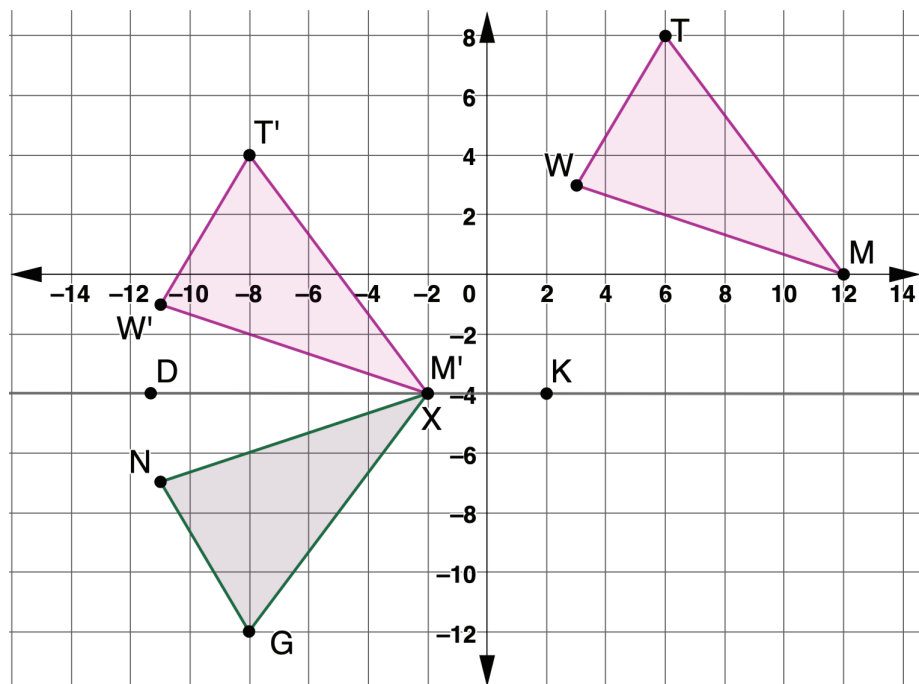
2. What do you wonder about the pattern or the final design?



5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Work with your partner on the following.

The diagram shows a sequence of transformations that, when applied to $\triangle TMW$, show that $\triangle TMW$ maps to $\triangle GXN$.



1. What are the transformations used to map $\triangle TMW$ to $\triangle GXN$?

2. Complete the table by determining which parts of $\triangle GXN$ correspond with the given part of $\triangle TMW$ after mapping $\triangle TMW$ to $\triangle GXN$.

$\triangle TMW$	$\triangle GXN$
$\overline{MT}$	
$\overline{MW}$	
$\overline{WT}$	
$\angle TMW$	
$\angle TWM$	
$\angle WTM$	



Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

3. Discuss with your partner how to determine if the transformations that were used to map $\triangle TMW$ to $\triangle GXN$ are rigid motions.
4. Complete the statement.

$\triangle TMW$ and $\triangle GXN$ are congruent because there exists

- ☐ 1
☐ 2
☐ 3

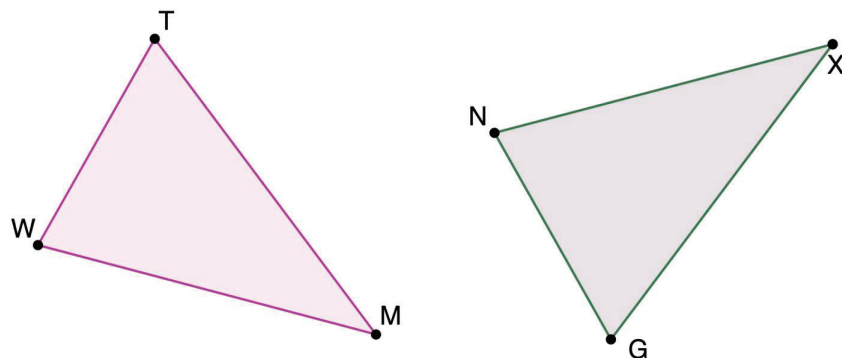
rigid motion(s) which map(s) $\triangle TMW$ to $\triangle GXN$.



5.4.3 Guided Instruction: CPCTC

Another way to show that triangles are congruent is with congruence postulates and theorems.

1. $\triangle TMW$ and $\triangle GXN$ are shown where $\angle GNX \cong \angle MWT$, $\angle NXG \cong \angle TMW$, and $\overline{XG} \cong \overline{MT}$.



- a. From the given information, what congruence postulate or theorem can be used to show that $\triangle TMW \cong \triangle GXN$?
- b. Discuss with your partner what could be concluded about the corresponding parts of the triangles if the congruence postulate or theorem is used to show the triangles are congruent.

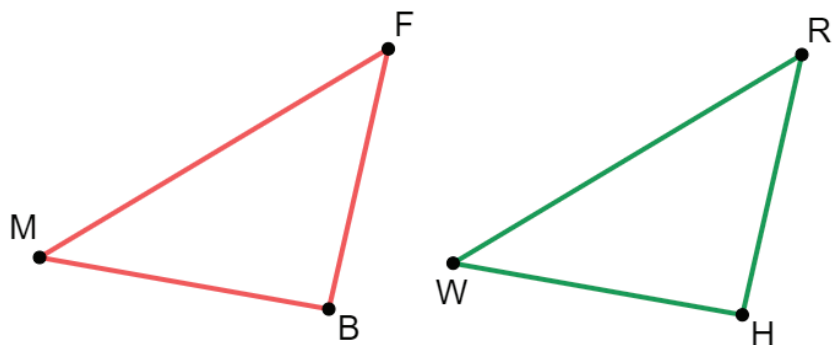


Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

There is a difference between naming two triangles and stating that two triangles are congruent. Order is important in a congruence statement, but is not necessary when only naming two triangles. The order used for congruence statements communicates which parts of the triangles correspond.

2. A diagram of two triangles is shown.

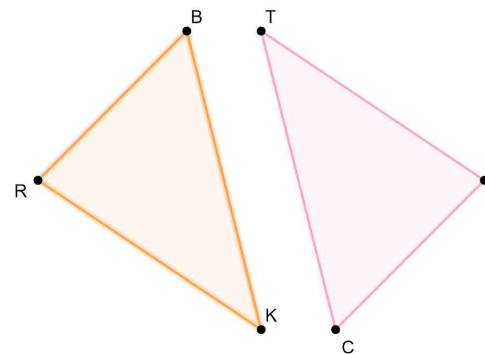


Naming	Congruence
ΔMFB and ΔHRW	$\Delta MFB \cong \Delta WRH$

Use the congruence statement to name all of the corresponding parts of the triangles.

ΔMFB	ΔWRH

3. Consider triangles BKR and CTY .



a. If $\Delta BKR \cong \Delta CTY$, mark the corresponding congruent sides and the corresponding congruent angles on the diagram.

b. Complete each congruence statement.

$$\overline{BK} \cong \underline{\hspace{2cm}} \quad \angle KBR \cong \angle \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} \cong \overline{CY} \quad \angle RKB \cong \angle \underline{\hspace{2cm}}$$

$$\overline{RK} \cong \underline{\hspace{2cm}} \quad \angle \underline{\hspace{2cm}} \cong \angle CYT$$

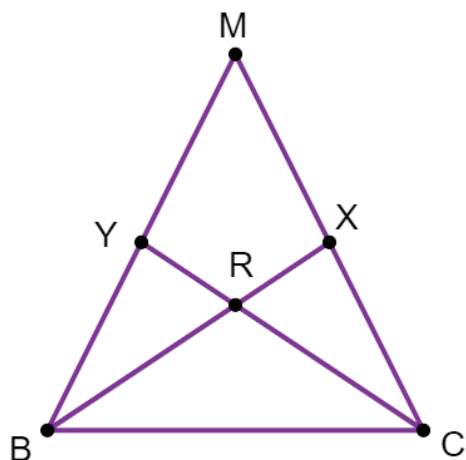
4. Consider two congruent triangles, ΔGRD and ΔFHW .

a. Write the congruence statement $\Delta GRD \cong \Delta FHW$ in two other ways.

b. Name the congruent sides for $\Delta GRD \cong \Delta FHW$.

c. Name the congruent angles for $\Delta GRD \cong \Delta FHW$.

5. Triangle MBC is shown.



a. How many different triangles are in $\triangle MBC$?

b. If $\triangle BRY \cong \triangle CRX$, then complete the congruence statement.

$$\overline{RC} \cong \underline{\hspace{2cm}} \quad \angle RYB \cong \underline{\hspace{2cm}}$$

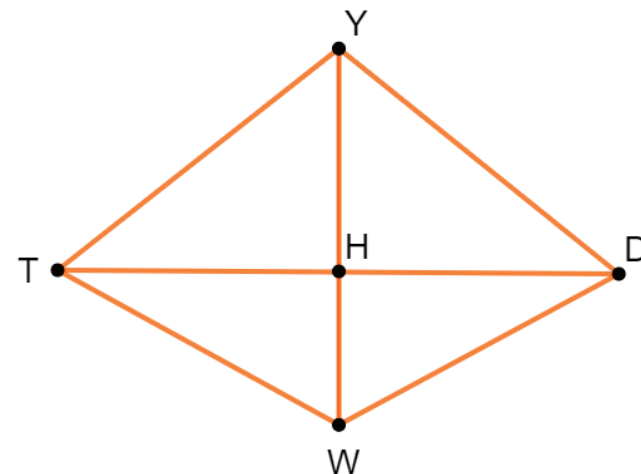
c. If $\triangle BXM \cong \triangle CYM$, then complete the congruence statement.

$$\overline{CY} \cong \underline{\hspace{2cm}} \quad \overline{CM} \cong \underline{\hspace{2cm}} \quad \angle MCY \cong \underline{\hspace{2cm}}$$



5.4.4 Guided Practice: Using CPCTC

Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



1. How many different triangles are in Quadrilateral $YDWT$?

2. If $\triangle DHY \cong \triangle THY$, then complete each congruence statement.

$$\overline{HD} \cong \underline{\hspace{2cm}} \quad \angle DYH \cong \underline{\hspace{2cm}} \quad \overline{TH} \cong \underline{\hspace{2cm}}$$

3. If $\triangle DHW \cong \triangle THW$, then complete each congruence statement.

$$\angle TWH \cong \underline{\hspace{2cm}} \quad \angle HTW \cong \underline{\hspace{2cm}} \quad \overline{TW} \cong \underline{\hspace{2cm}}$$

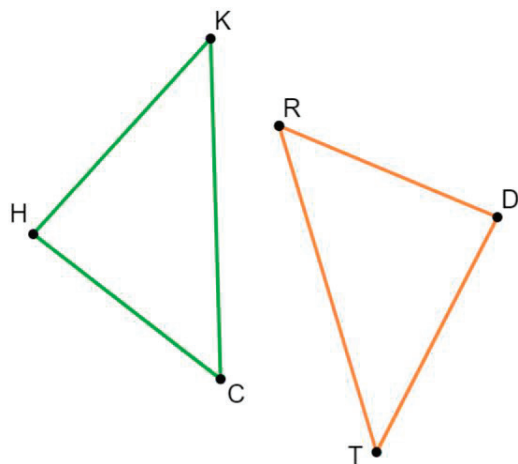
4. If $\triangle YTW \cong \triangle YDW$, then complete each congruence statement.

$$\angle WTY \cong \underline{\hspace{2cm}} \quad \overline{DW} \cong \underline{\hspace{2cm}} \quad \overline{YW} \cong \underline{\hspace{2cm}}$$



5.4.5 Wrap-Up: Cool Down

The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle DTR$.



- Complete each congruence statement.

$$\overline{KC} \cong \underline{\hspace{2cm}}$$

$$\overline{CH} \cong \underline{\hspace{2cm}}$$

$$\angle RTD \cong \underline{\hspace{2cm}}$$

$$\angle RDT \cong \underline{\hspace{2cm}}$$

- Becky told her teacher that there was a typo in the first sentence. She circled the typo.

The diagram shows $\triangle HKC$ and $\triangle DTR$ where $\triangle HKC \cong \triangle DTR$.

Write a note to Becky explaining why this is NOT a typo.

Unit 5, Lesson 5: Drawing Conclusions from Given Information



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can draw and mark figures using given information.
- ☐ I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from given information.



5.5.1 Warm-Up: Working Through Challenges

Teri takes dance in school, but learning how to tap dance was a new and overwhelming experience. She said that after her teacher introduced tap, “It was really hard for me to learn ... It just looked like a lot of work and looked like it took a lot of time to learn, and it was really high tempo.”

1. What is something that felt overwhelming the first time you were learning it?

Teri’s teacher slowed down the moves to show her the structure and recommended that she practice each step before trying to put it all together. She connected this experience to learning other things that are difficult, saying, “The more you practice a certain skill that you need, the better you become at it, and the easier it becomes.”

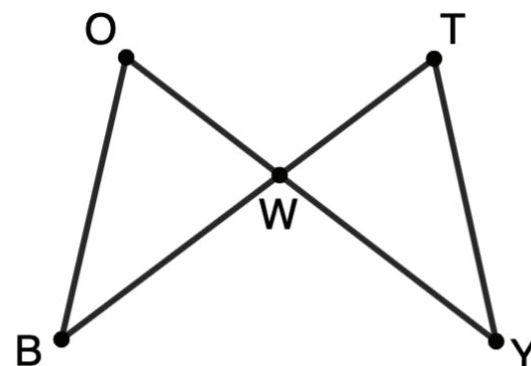
2. Describe a time when practicing something that was challenging for you helped you become better at it.



5.5.2 Collaboration: Using Given Information to Mark Figures

When given information about geometric figures, it is helpful to mark the figures.

1. Triangles OBW and WTY are shown.



Josue and Abby marked the figure in different ways given $\overline{OW} \cong \overline{TW}$, $\overline{OB} \cong \overline{TY}$, and $\angle OBW \cong \angle TYW$ as shown.

Josue's Markings	Abby's Markings

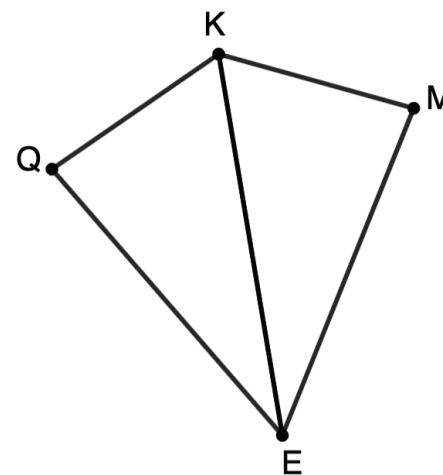
- a. Discuss with your partner what you notice about how Josue and Abby each marked the figure.

- b. Complete the table to describe similarities and differences in the way each student marked the figure.

How were their markings similar?	How were their markings different?

- c. Josue wondered if there was enough information given to conclude $\triangle OWB \cong \triangle TWY$. Work with your partner to explain whether there is enough information to conclude $\triangle OWB \cong \triangle TWY$.
- d. Abby said she does not think there is enough given information to conclude $\triangle OWB \cong \triangle TWY$, but she said there is enough information to conclude $\angle OWB \cong \angle TWY$. What definition is Abby using to draw this conclusion?
- e. Add markings to Abby's figure to represent her conclusion that $\angle OWB \cong \angle TWY$.
- f. Work with your partner to explain whether there is now enough information to conclude $\triangle OWB \cong \triangle TWY$.

2. Mark polygon $KQEM$ using the given information.
 $\overline{EQ} \cong \overline{EM}$ and $\angle KEQ \cong \angle KEM$



3. Compare your markings with your partner. Make any corrections if necessary.



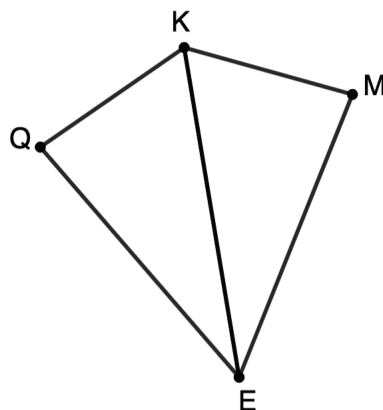
5.5.3 Guided Instruction: Drawing Conclusions from Given Information

Given information, definitions, properties, postulates, and theorems can be used to draw conclusions about figures.

1. What are some definitions, properties, postulates, and theorems related to triangles and angles learned so far this year?

Definitions	
Properties	
Theorems and Postulates	

Polygon $KQEM$, where $\overline{QK} \cong \overline{KM}$ and $\overline{EQ} \cong \overline{EM}$, is shown. $\overline{KE}$ is drawn inside polygon $KQEM$.



2. Mark the figure using the given information.
3. Discuss any conclusions that can be drawn about the figure with the information given.

Using the given information, Hilary concludes $\triangle QEK \cong \triangle MEK$.

4. Complete the statement to explain how Hilary is able to draw this conclusion.

Hilary noticed ☐ 1
☐ 2
☐ 3 corresponding ☐ side(s)
☐ angle(s) of

$\triangle QEK$ and $\triangle MEK$ are congruent based on the given information.

Both triangles also share a common ☐ side,
☐ angle,

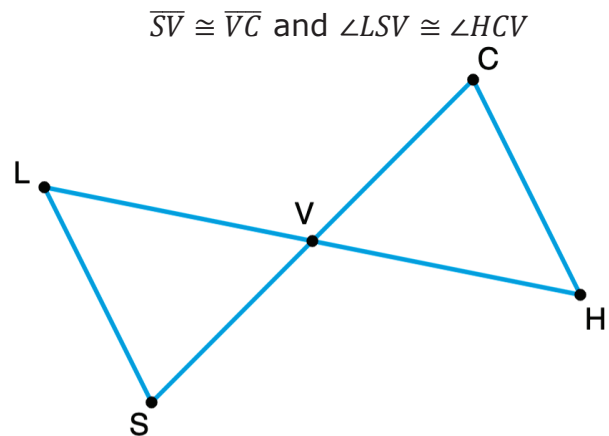
which is congruent to itself because of the

☐ reflexive
☐ transitive property.

She can conclude $\triangle QEK \cong \triangle MEK$ because of

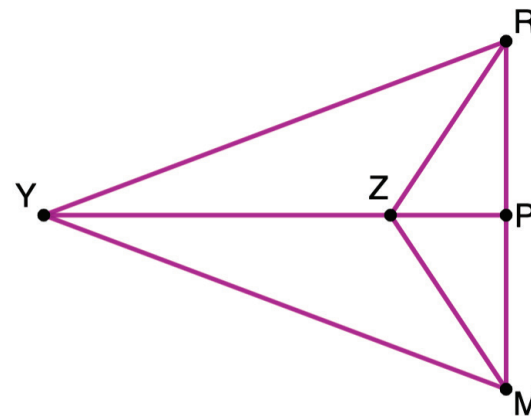
☐ SSS.
☐ SAS.
☐ ASA.
☐ AAS.
☐ HL.

5. Triangle LVS and triangle CVH are shown.



- What conclusions can be drawn from the given information?
- What congruence postulate or theorem is the reason these conclusions can be made?
- Julia says she can conclude $\overline{CH} \cong \overline{SL}$ and $\angle SLV \cong \angle CHV$. Explain why Julia is able to draw these conclusions.

6. Triangle RMZ is shown. $\angle YMZ \cong \angle YRZ$ and $\angle RZY \cong \angle MZY$.



Work with your partner to complete the statements based on the given information.

- I can conclude ...
- I know this because ...

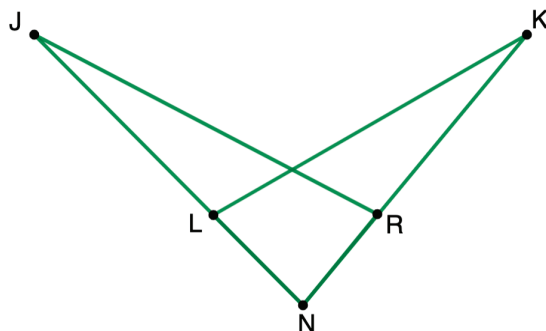


5.5.4 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

Dalton saw the following question with the polygon shown in an old geometry textbook.

"If $\angle NKL \cong \angle NJR$ and $\overline{KL} \cong \overline{JR}$, what else do you know so that you can show that $\triangle RJN \cong \triangle LKN$ using ASA?"



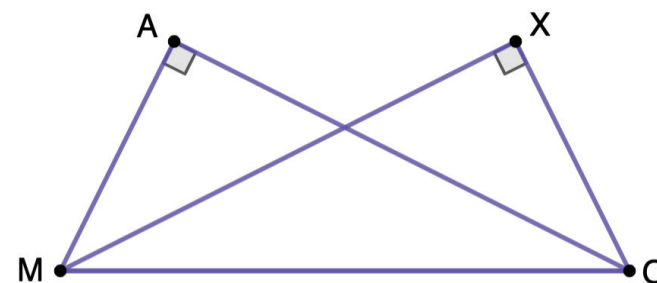
He knew that both triangles share $\angle N$. Therefore, by knowing that $\angle N \cong \angle N$ by the reflexive property of congruence, he concluded that he could use ASA to show that $\triangle RJN \cong \triangle LKN$.

1. Draw the triangles so that they do not overlap. Then, mark the triangles with the information given from the textbook.
2. Explain why Dalton can or cannot use ASA to show that $\triangle RJN \cong \triangle LKN$.



5.5.5 Try It: Drawing Conclusions from Given Information

1. Triangle MAO and triangle XOM are shown, where $\angle MAO$ and $\angle OXM$ are right angles and $\overline{MA} \cong \overline{XO}$.



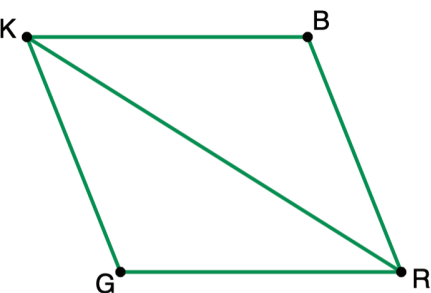
Complete each statement based on the given information.

- a. I can conclude that ...
- b. I know this because ...



5.5.6 Wrap-Up: Ticket Out the Door

Quadrilateral $BRGK$ is shown.



Complete each statement based on the given information.

1. Given that $\overline{KB} \cong \overline{GR}$ and $\overline{KG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
2. Given that $\overline{KB} \cong \overline{KG}$ and $\overline{RG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
3. Given that $\overline{KG} \parallel \overline{BR}$ and $\overline{KB} \parallel \overline{GR}$,
 - a. I can conclude that ...
 - b. I know this because ...

Unit 5, Lesson 6: Proving Triangles Congruent



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.



5.6.1 Warm-Up: Cheese Balls

The image shows cheese balls arranged on a baking sheet with a white plate on top. There are thirteen cheese balls on the plate.



1. Estimate how many cheese balls will fit on the plate.
2. Explain how you determined your estimate.



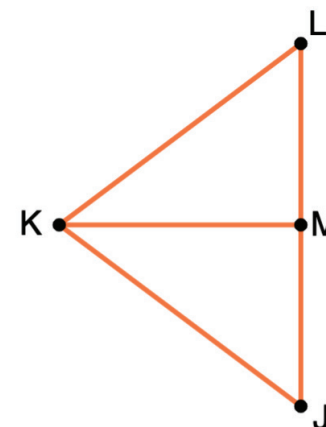
5.6.2 Guided Instruction: Proving Triangles Congruent

When completing a proof to show two triangles are congruent, it is helpful to start by considering conclusions that can be drawn from the given information.

1. Triangle KLJ is shown with some given information.

Given: $\overline{KM}$ bisects $\overline{LJ}$
 $\overline{KL} \cong \overline{KJ}$

- a. What conclusion can be drawn from the given statement, " $\overline{KM}$ bisects $\overline{LJ}$ "?
- b. Explain why this conclusion can be drawn.
- c. Mark the figure based on the given information.
- d. What other conclusion(s) can be drawn from the information given?

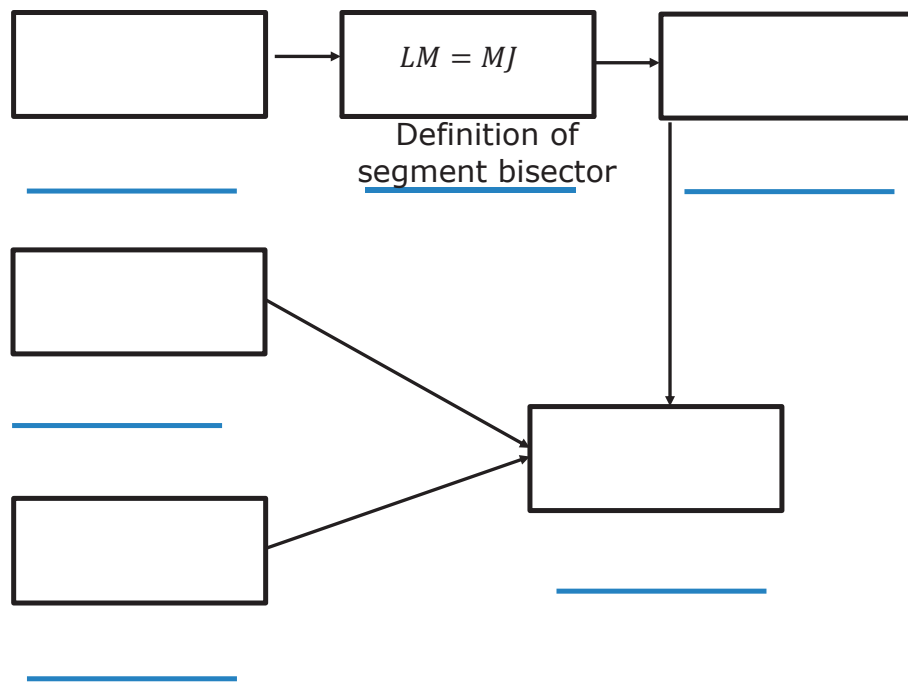


e. Explain why this conclusion can be drawn.

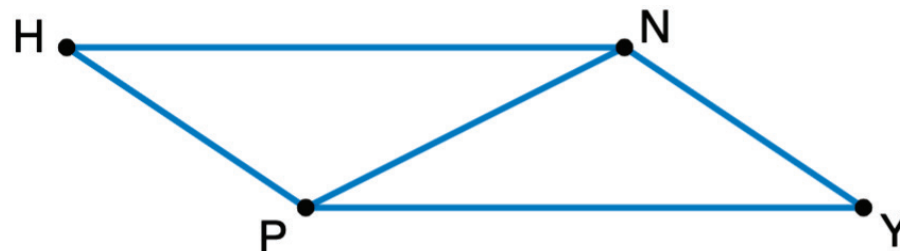
f. Use the conclusions drawn about $\triangle KIJ$ from the information given, along with the figure, to complete the flow chart proof.

Given: $\overline{KM}$ bisects $\overline{IJ}$
 $\overline{KL} \cong \overline{KJ}$

Prove: $\triangle KIM \cong \triangle KJM$



2. Polygon $HPYN$ is shown.



a. Mark the diagram to show the following given information.

Given: $\overline{HP} \cong \overline{YN}$ and $\overline{HP} \parallel \overline{YN}$

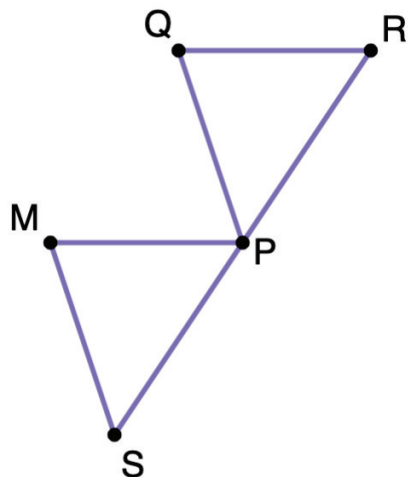
b. Discuss how you determined which markings to make with your partner.

c. Which congruence theorem or postulate should be used to prove $\triangle PHN \cong \triangle NYP$?

d. Complete the two-column proof.

Statement	Reason
1. $\overline{HP} \cong \overline{YN}$	1. Given
2. $\overline{HP} \parallel \overline{YN}$	2.
3.	3.
4.	4.
5. $\triangle PHN \cong \triangle NYP$	5.

3. Triangles MSP and QPR are shown.



Given: $\overline{QR} \parallel \overline{MP}$
 $\angle SMP \cong \angle PQR$
 P is the midpoint of $\overline{SR}$

Prove: $\triangle SPM \cong \triangle PRQ$

- Mark the diagram to show the given information.
- Complete the paragraph proof.

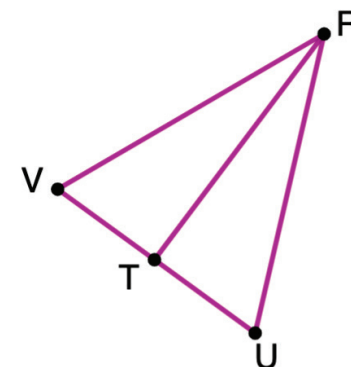
It is given that P is the midpoint of $\overline{SR}$, so _____
 by _____. It is given that $\overline{QR} \parallel \overline{MP}$,
 so $\angle _____ \cong \angle _____$ by _____.
 It is also given that _____. Therefore,
 $\triangle SPM \cong \triangle PRQ$ by _____.



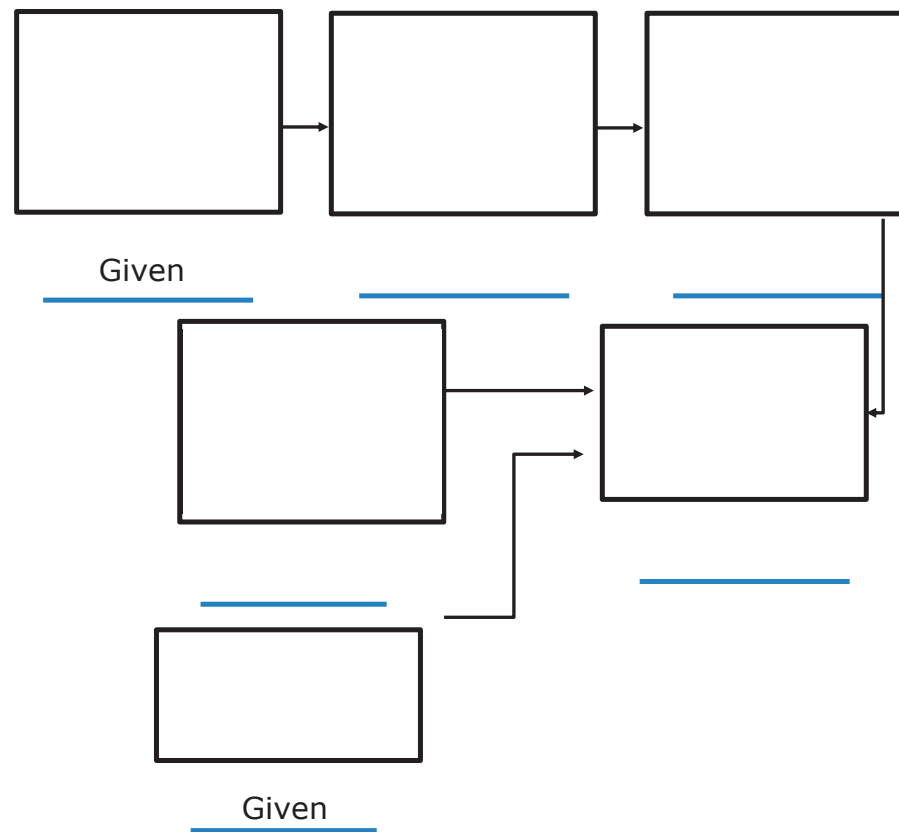
5.6.3 Guided Practice: Proving Triangles Congruent

1. Triangle VRU is shown.

Given: $\overline{UR} \cong \overline{VR}$
 $\overline{RT} \perp \overline{UV}$
 Prove: $\triangle RTV \cong \triangle RTU$



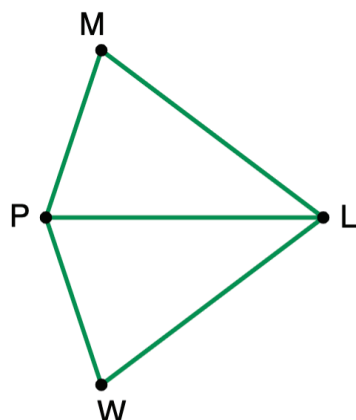
Complete the flow chart proof.





5.6.4 Your Turn!

1. Polygon $MLWP$ is shown.



Given: $\overline{LP}$ bisects $\angle MLW$
 $\angle LPM \cong \angle LPW$

Prove: $\triangle WPL \cong \triangle MPL$

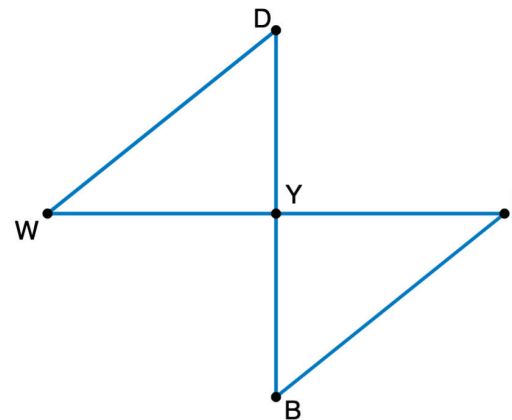
Complete the paragraph proof.

It is given that $\overline{LP}$ bisects $\angle MLW$, so _____ by definition of _____. Also, it is given that _____. _____ $\cong$ _____ by _____. Therefore, $\triangle WPL \cong \triangle MPL$ by _____.



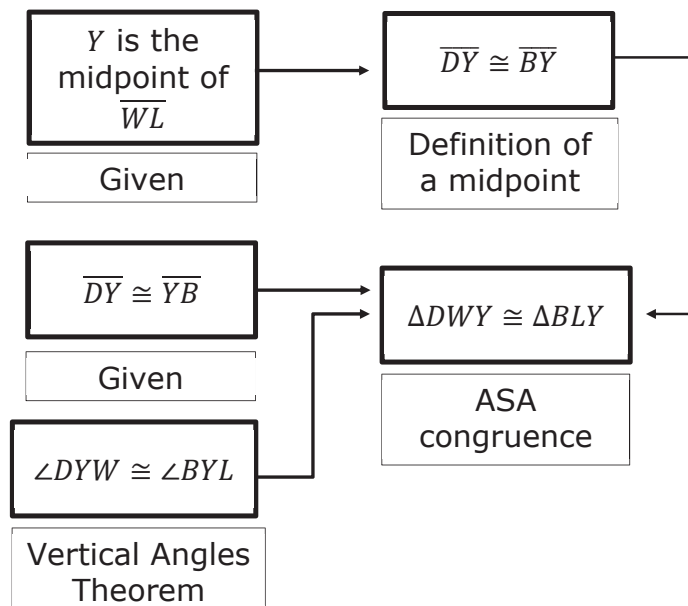
5.6.5 Wrap-Up: Error Analysis

1. The flow chart proof has two errors. Find the errors and correct them.



Given: Y is the midpoint of $\overline{WL}$
 $\overline{DY} \cong \overline{YB}$

Prove: $\triangle DWY \cong \triangle BLY$



Unit 5, Lesson H1: Definition of Congruence in Terms of Rigid Transformations



Benchmark

MA.912.GR.2.7 Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations.



Learning Target

- ☐ I can use the definition of congruence to justify that SSS congruence, SAS congruence, ASA congruence, and AAS congruence are sufficient to show that two triangles are congruent.



5.H1.1 Warm-Up: Geometry as an Animal

1. If geometry were an animal, it would be a ...

because ...

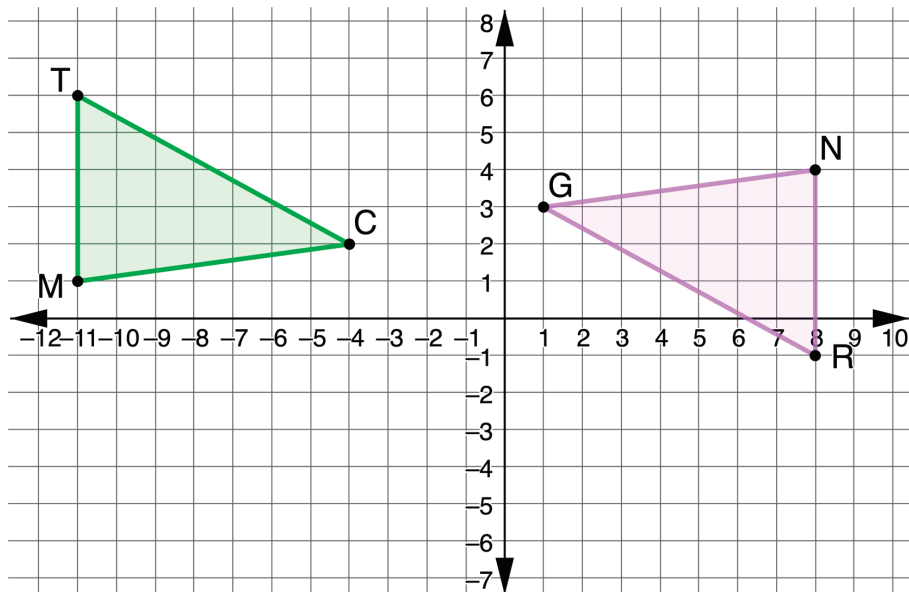
2. It would live in ...

because ...



5.H1.2 Exploration: Justifying Congruence

1. Triangles TCM and RGN are shown, where $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$.



- a. Discuss with your partner what can be concluded from the given information.

- b. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

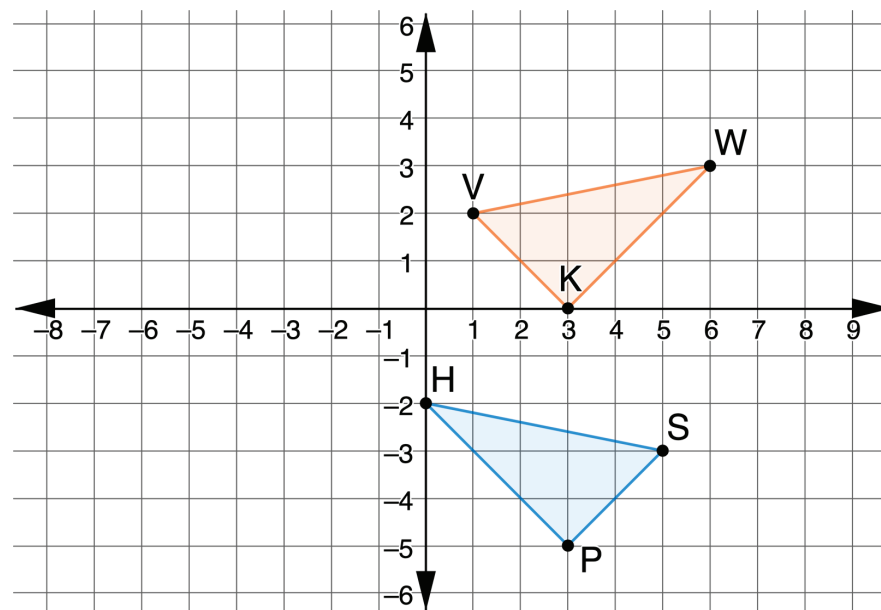
- c. What rigid motions can be applied so that $\triangle TCM$ is mapped to $\triangle RGN$?
- d. Discuss with your partner what can be concluded from the rigid motions.

- e. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

- f. Discuss with your partner why showing that the triangles are congruent using the definition of congruence in terms of rigid transformations shows that SAS congruence is sufficient to prove the triangles are congruent. Summarize your discussion.

2. Triangles VWK and SHP are shown. Assume $\overline{VK} \cong \overline{SP}$, $\overline{WK} \cong \overline{PH}$, and $\overline{KW} \cong \overline{KH}$.



- a. Explain how you could use patty paper to verify $\triangle VWK \cong \triangle SHP$.
- b. Write a series of transformations that will map $\triangle VWK$ onto $\triangle SHP$.

- c. Show each on the coordinate grid.
- Draw $\Delta V'W'K'$, the result of the first transformation
 - Map the vertices of ΔVWK to $\Delta V'W'K'$
 - Map the vertices of $\Delta V'W'K'$ to ΔSHP

d. Complete the statements.

The outcome of mapping the transformations of

ΔVWK to $\Delta V'W'K'$ then from $\Delta V'W'K'$ to Δ _____ shows

that V' coincides with $\begin{matrix} \circ H, \\ \circ S, \\ \circ P, \end{matrix}$ W' coincides

with $\begin{matrix} \circ H, \\ \circ S, \\ \circ P, \end{matrix}$ and K' coincides with $\begin{matrix} \circ H. \\ \circ S. \\ \circ P. \end{matrix}$

Because the definition of $\begin{matrix} \circ \text{congruence} \\ \circ \text{similarity} \end{matrix}$

in terms of rigid motions preserves distance and

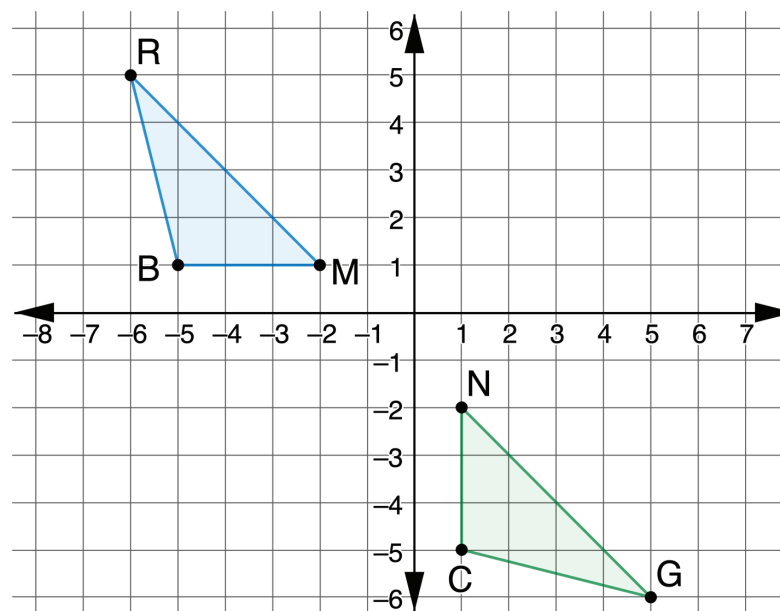
angles, triangle VWK is $\begin{matrix} \circ \text{congruent} \\ \circ \text{similar} \end{matrix}$ to triangle

_____. Since $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$

was given, $\begin{matrix} \circ \text{AAS} \\ \circ \text{ASA} \\ \circ \text{SAS} \\ \circ \text{SSS} \end{matrix}$ is sufficient to show triangles

are congruent.

3. Triangles RMB and GNC are shown, where $\overline{RM} \cong \overline{GN}$, $\angle BRM \cong \angle CGN$, and $\angle BMR \cong \angle CNG$.



Adriana knows that there is a series of rigid motions that maps ΔRMB onto ΔGNC .

a. Select all true statements after the transformations.

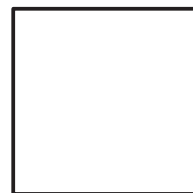
- ☐ Line segment RB coincides with line segment GN .
- ☐ Line segment BM coincides with line segment CN .
- ☐ Line segment CG coincides with line segment BR .
- ☐ Angle NCG coincides with angle RMB .
- ☐ Angle RBM coincides with angle GNC .

- b. Explain how the given information and the definition of congruence in terms of rigid motions shows that $\triangle RMB \cong \triangle GNC$.

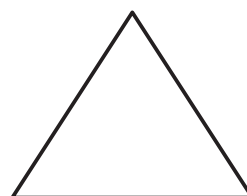


5.H1.3 Wrap-Up: Reflection

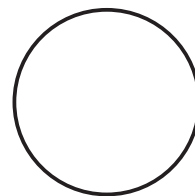
1. What is something that is now “squared away” in your mind about the definition of congruence in terms of rigid transformations?



2. What are three “points” that are important for showing triangle congruence using the definition of congruence in terms of rigid transformations?



3. What is something that is still “circling” your brain about the definition of congruence in terms of rigid transformations?



Unit 5, Lesson 7: Corresponding Parts of Congruent Triangles are Congruent



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I understand how to use corresponding parts of congruent triangles in a proof.
- ☐ I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.



5.7.1 Warm-Up: Brain Teaser

Brain teasers like this have been traced back to medieval times. Here's one with a Florida twist.

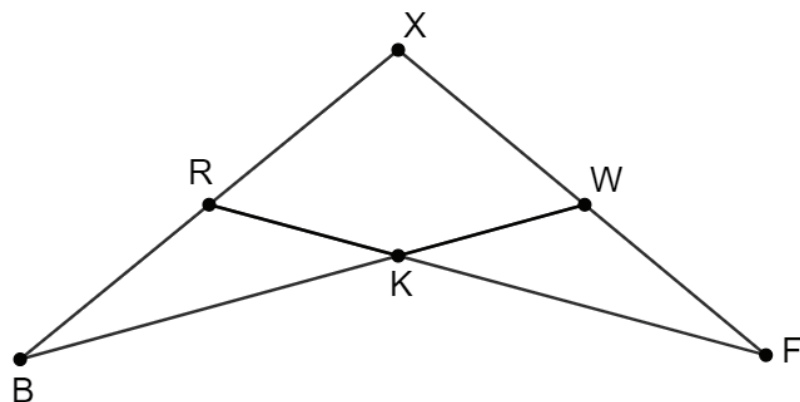
1. A man has to take a panther, a raccoon, and some strawberries across St. John's river. His rowboat has enough room for the man, plus either the panther or the raccoon or the strawberries. If he takes the strawberries with him, the panther will eat the raccoon. If he takes the panther, the raccoon will eat the strawberries. Only when the man is present are the raccoon and the strawberries safe from their enemies. All the same, the man successfully carries panther, raccoon, and strawberries across the river. How?



5.7.2 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

Polygon $XFKB$ is shown.



1. How many triangles are in polygon $XFKB$?

2. Draw $\triangle XBW$ and $\triangle XFR$ separately. Label where point K is on each triangle.

3. Discuss with your partner if triangles $\triangle XBW$ and $\triangle XFR$ have any common sides or angles. If so, name the common part(s).

4. Explain why knowing if $\triangle XBW$ and $\triangle XFR$ have any common sides or angles could be helpful in a proof.

5. In your drawing of $\triangle XBW$ and $\triangle XFR$ as separate triangles, mark the common part(s) as congruent.

6. What property justifies that something is equal to itself?

7. Mark the given information in your diagram.

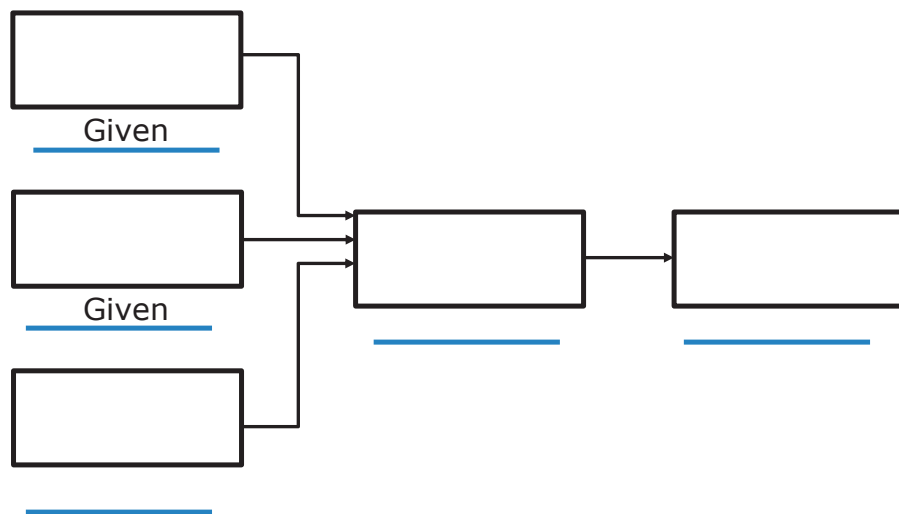
Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$

8. What can you conclude about $\triangle XBW$ and $\triangle XFR$ from the information you marked on your drawing?

9. Complete the flowchart proof for polygon $XFKB$.

Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$

Prove: $\angle XBK \cong \angle XFK$



10. Check your proof with another group and if necessary, make adjustments.



5.7.3 Guided Instruction: Proofs Using CPCTC

Once two triangles have been proven to be congruent, then CPCTC states that any other corresponding parts of the triangles are also congruent.

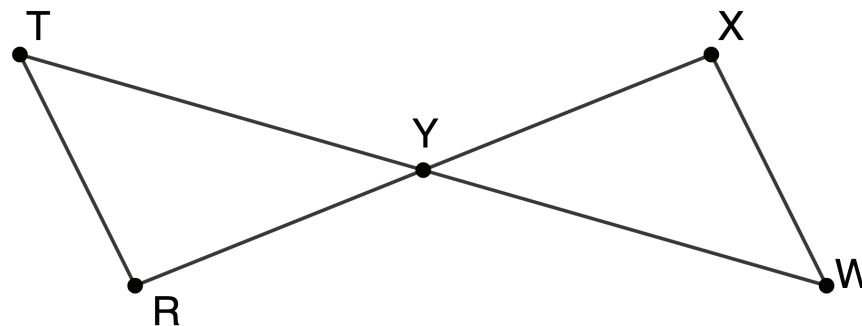


Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

In the flowchart proof, proving that $\angle XBK \cong \angle XFK$ first required proving that $\triangle XBW$ and $\triangle XFR$ are congruent.

1. Triangles TYR and XYW are shown.



a. Mark the diagram to show the following given information.

Given: Y is the midpoint of $\overline{TW}$. Y is the midpoint of $\overline{RX}$.

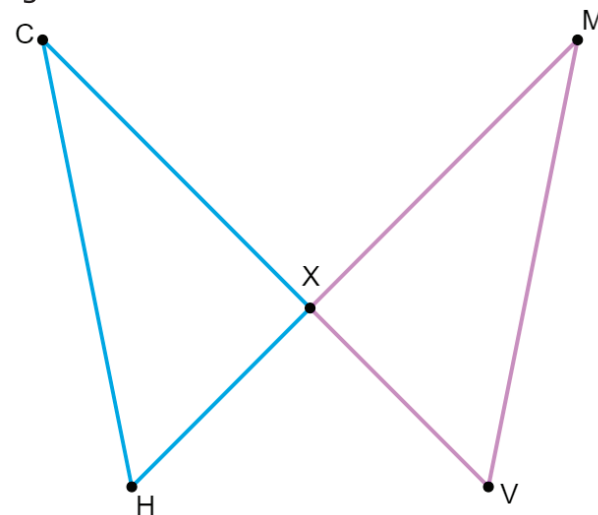
- b. What does “Y is the midpoint of $\overline{WT}$ ” allow you to conclude about the diagram?
- c. What does “Y is the midpoint of $\overline{RX}$ ” allow you to conclude about the diagram?
- d. Discuss with your partner if there is any other information about the two triangles that can be determined from the diagram. Write a summary of your discussion.
- e. Work with your partner to complete the paragraph proof.

Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
 Prove: $\overline{TR} \cong \overline{XW}$

It is given that Y is the midpoint of $\overline{WT}$, so _____ by _____.
 Also, it is given that Y is the midpoint of $\overline{RX}$, so _____ by _____.
 _____ by _____. Therefore by _____,
 $\triangle \underline{\hspace{1cm}} \cong \triangle \underline{\hspace{1cm}}$ so by _____,
 $\overline{TR} \cong \overline{XW}$.



2. Triangles CXH and XMV are shown.



Given: $\overline{CV} \perp \overline{HM}$, $\overline{HX} \cong \overline{XV}$, $\overline{HC} \cong \overline{MV}$
 Prove: $\overline{XC} \cong \overline{XM}$

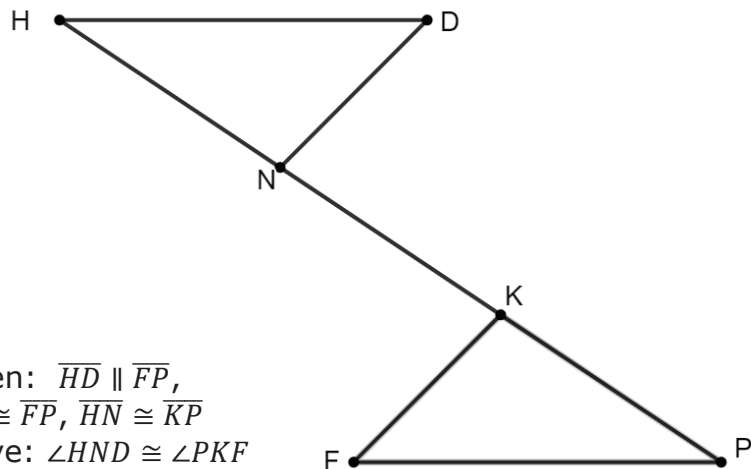
- a. Mark the diagram to show the given information.
- b. Which congruence theorem or postulate do you think should be used?
- c. Complete the two-column proof.

Statement	Reason
1. $\overline{CV} \perp \overline{HM}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7. $\overline{XC} \cong \overline{XM}$	7.



5.7.4 Guided Practice: Proofs Using CPCTC

1. A diagram of triangles NDH and PKF with $\overline{NK}$ is shown.



Given: $\overline{HD} \parallel \overline{FP}$,
 $\overline{HD} \cong \overline{FP}$, $\overline{HN} \cong \overline{KP}$

Prove: $\angle HND \cong \angle PKF$

Complete the paragraph proof.

It is given that $\overline{HD} \parallel \overline{FP}$, so by _____,

$\angle \underline{\hspace{1cm}} \cong \angle \underline{\hspace{1cm}}$. Since it is given that $\overline{HD} \cong \overline{FP}$ and

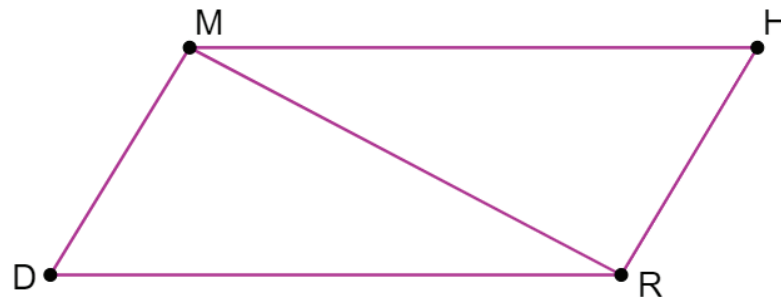
$\overline{HN} \cong \overline{KP}$, $\Delta \underline{\hspace{1cm}} \cong \Delta \underline{\hspace{1cm}}$ by _____. Therefore,

$\angle HND \cong \angle PKF$ by _____.



5.7.5 Your Turn!

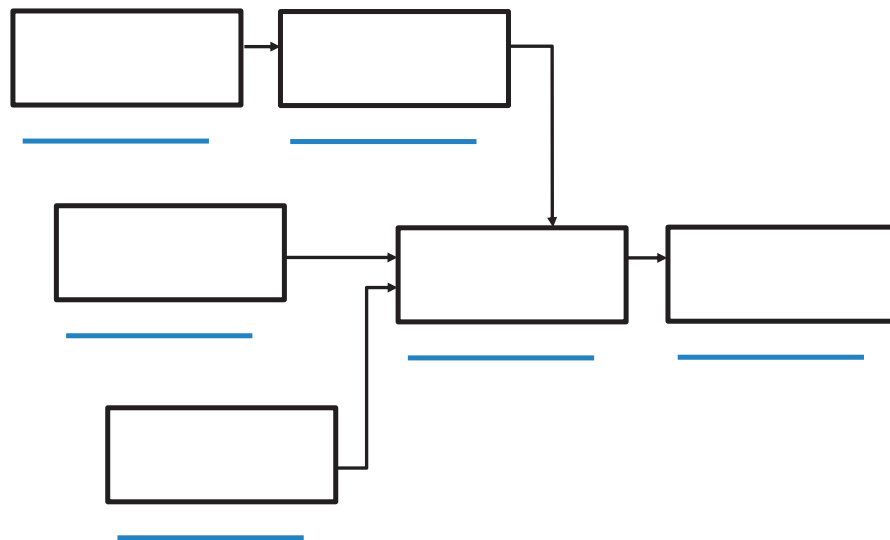
1. Quadrilateral $MHRD$ is shown.



Given: $\overline{DM} \parallel \overline{RH}$, $\angle RDM \cong \angle RHM$

Prove: $\overline{DM} \cong \overline{RH}$

Complete the flowchart proof.





5.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I understand how to use corresponding parts of congruent triangles in a proof.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
--------------------------------------	---	--



2. I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
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Unit 5, Lesson 8: The Proof is in the Bag!



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.
- ☐ I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

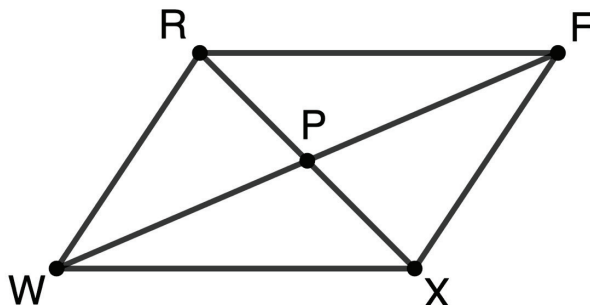


5.8.1 Warm-Up: Bell Work

Polygon $RFXW$
where diagonals
 $\overline{RX}$ and $\overline{FW}$ bisect
each other at
point P is shown.

Given: $\overline{RX}$ and
 $\overline{FW}$ bisect each
other at point P .

Prove: $\triangle RPF \cong \triangle XPW$



1. Complete the two-column proof.

Statement	Reason
1. $\overline{RX}$ and $\overline{FW}$ bisect each other at point P	1. Given
2.	2. Definition of _____ bisector
3. $\overline{RP} \cong \underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}} \cong \overline{WP}$	3. Definition of _____
4.	4.
5. $\triangle RPF \cong \triangle XPW$	5.



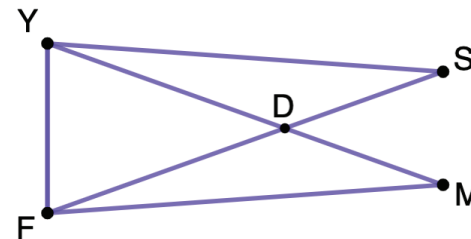
5.8.2 Activity: It's in the Bag!

With your group, complete each proof using the statement and reason cards provided.

Proof A

Given: $\angle YSD \cong \angle FMD$
 $\triangle YDF$ is
isosceles with
base $\overline{YF}$

Prove: $\overline{FM} \cong \overline{YS}$

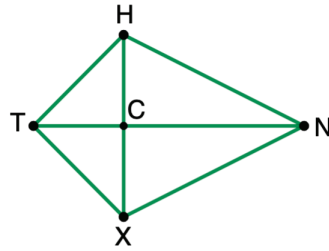


Statement	Reason

Proof B

Given: $\overline{TN}$ bisects HTX
 $\overline{HT} \cong \overline{XT}$

Prove: $\overline{TN}$ bisects $\angle HNX$

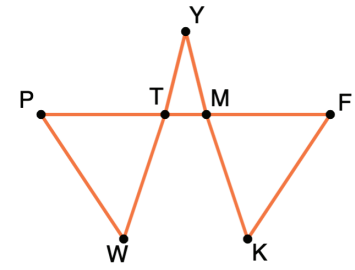


Statement	Reason

Proof C

Given: $\angle WPT \cong \angle KFM$
 $\overline{PW} \cong \overline{FK}$
 $\angle YMT \cong \angle YTM$

Prove: $\triangle PWT \cong \triangle FKM$

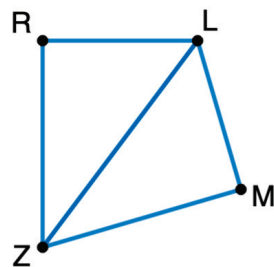


Statement	Reason

Proof D

Given: $\overline{LM} \cong \overline{LR}$
 $\overline{LR} \perp \overline{RZ}$
 $\overline{LM} \perp \overline{MZ}$

Prove: $\angle MZL \cong \angle RZL$

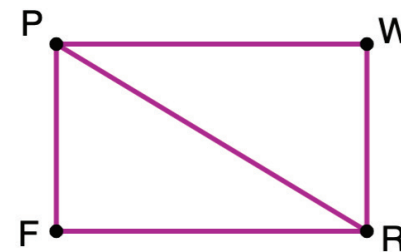


Statement	Reason

Proof E

Given: $\overline{PF} \cong \overline{WR}$
 $\overline{PW} \cong \overline{FR}$

Prove: $\overline{PF} \parallel \overline{WR}$

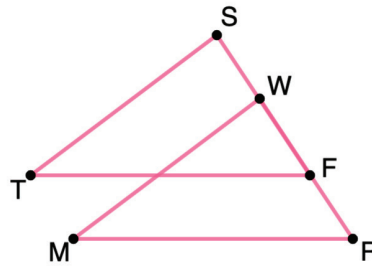


Statement	Reason

Proof F

Given: $\overline{RF} \cong \overline{SW}$
 $\angle TSF \cong \angle MWR$
 $\overline{FT} \parallel \overline{MR}$

Prove: $\triangle TSF \cong \triangle MWR$



Statement	Reason



5.8.3 Wrap-Up: Cool Down

- Write an explanation to a classmate who might have missed today's lesson, in your own words, about writing proofs.

Unit 5, Lesson 9: Congruence in Two-Dimensional Figures – Part 1



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



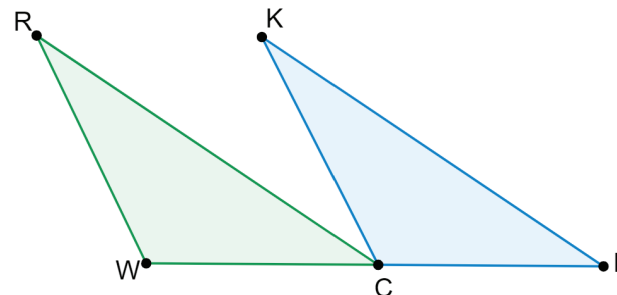
Learning Target

- ☐ I can solve mathematical problems involving congruence.



5.9.1 Warm-Up: Bell Work

Triangles RCW and KHC are shown.



Given: $RW \parallel KC$, C is the midpoint of $\overline{WH}$, $\overline{WR} \cong \overline{CK}$

Prove: $\angle WRC \cong \angle CKH$

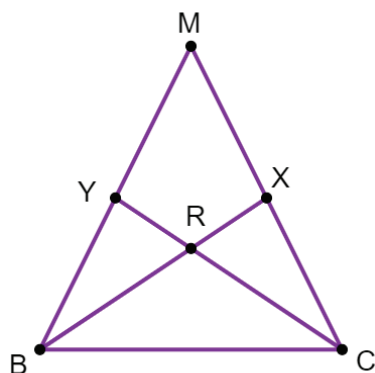
1. Complete the missing statements and reasons in the two-column proof.

Statement	Reason
1. C is the midpoint of $\overline{WH}$	1. Given
2.	2. Definition of midpoint
3. $RW \parallel KC$	3. Given
4.	4.
5.	5. Given
6.	6.
7. $\angle WRC \cong \angle CKH$	7.



5.9.2 Collaboration: Using CPCTC to Determine Measures

Triangle MBC is shown where $MX = 3.6$, $MB = 7.2$, $RX = 2$, $BC = 6.5$, $RB = 3.9$, $m\angle YBR = 29^\circ$, $m\angle CXR = 83^\circ$, and $m\angle MCB = 63^\circ$.



Work with your partner to complete the following.

- How many different triangles are in $\triangle MBC$?
- If $\triangle BRY \cong \triangle CRX$, then determine the measure of each.
 $RC = \underline{\hspace{2cm}}$ $m\angle RYB = \underline{\hspace{2cm}}$
- If $\triangle BXM \cong \triangle CYM$, then determine the measure of each.
 $CY = \underline{\hspace{2cm}}$ $CM = \underline{\hspace{2cm}}$ $m\angle MCY = \underline{\hspace{2cm}}$



5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

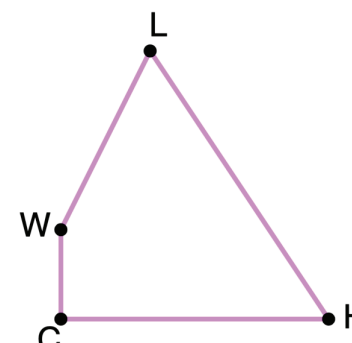
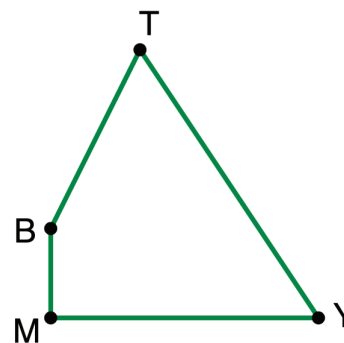
Any polygon that can be mapped to another polygon using rigid transformations shows that all of the parts of the polygons are congruent. This understanding can then be applied to find measurements of the two congruent polygons.



Corresponding Parts of Congruent Polygons are Congruent

If two polygons are congruent, then all corresponding parts of the polygons are congruent.

- Congruent quadrilaterals $TYMB$ and $LHCW$ are shown.



Some measurements of the angles in quadrilaterals $TYMB$ and $LHCW$ are given.

$$m\angle YMB = 90^\circ, m\angle WLH = 60^\circ, m\angle MBT + m\angle CHL = 210^\circ, \\ m\angle BTY = (2x - 4)^\circ, m\angle MYT = (x + 25)^\circ$$

- a. Determine each.

$$m\angle BTY$$

$$m\angle CWL$$

Some measurements of the sides of quadrilaterals $TYMB$ and $LHCW$ are given.

$$HC = 5k, \quad LW = 4k - 0.5, \quad YT = 6k, \quad \text{and} \quad BM = k + 1.$$

- b. Determine the value of k if the perimeter of quadrilateral $TYMB$ is 24.5.
- c. What is the length of $\overline{BT}$?

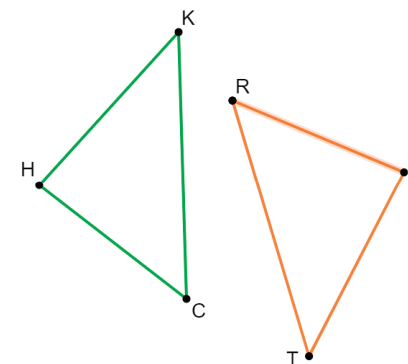
2. The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$.

Some measurements of the sides are given.

$$KH = 5y - 2,$$

$$DR = 6y - 5, \quad \text{and}$$

$$TD = 4y + 3$$



Arthur used the information to determine the length of KH . His work is shown.

Arthur's Work

$$5y - 2 = 6y - 5$$

$$-2 = y - 5$$

$$3 = y$$

$$KH = 5(3) - 2$$

$$KH = 13$$

- a. Find Arthur's error.
- b. Help Arthur understand and learn from his mistake. Write a brief note to Arthur explaining what his mistake was and how he can learn from the mistake.

Means

I

Start

To

Acquire

Knowledge

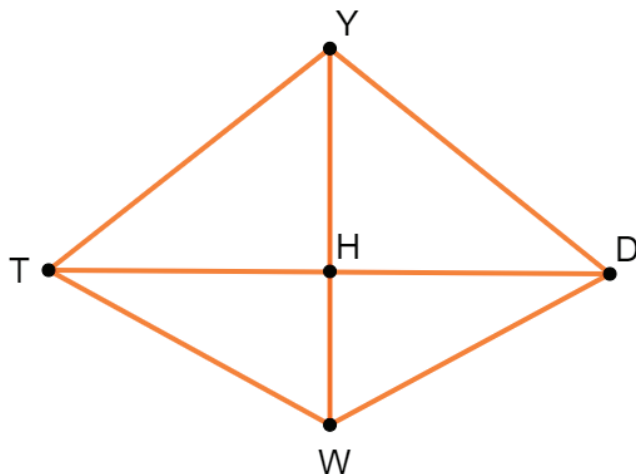
Experience

Skills



5.9.4 Guided Practice: Corresponding Parts

1. Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



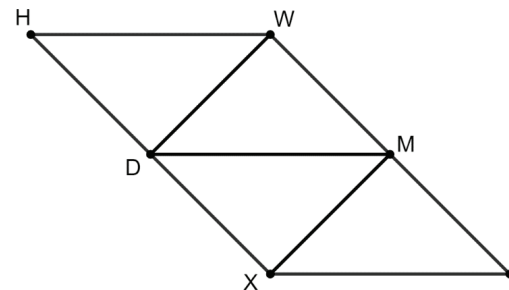
- a. How many different triangles are in Quadrilateral $YDWT$?
- b. If $\triangle DHY \cong \triangle THY$, $YD = \sqrt{130}$, $YH = 7$, and $m\angle HTY = 38^\circ$, determine the following measures.

$$HD = \underline{\hspace{2cm}} \quad m\angle DYH = \underline{\hspace{2cm}} \quad TH = \underline{\hspace{2cm}}$$

- c. If $\triangle DHW \cong \triangle THW$, $WH = 5$, and $m\angle HWD = 61^\circ$, use these and all other known measurements to determine the following measures.

$$m\angle TWH = \underline{\hspace{2cm}} \quad m\angle HTW = \underline{\hspace{2cm}} \quad TW = \underline{\hspace{2cm}}$$

2. Quadrilateral $HWLX$ is shown.



$HWMX \cong LXDW$ and $\triangle XMD \cong \triangle WDM$. The perimeter of $HWMX$ is 40.5.

- a. Use the given measurements to find the value of x .

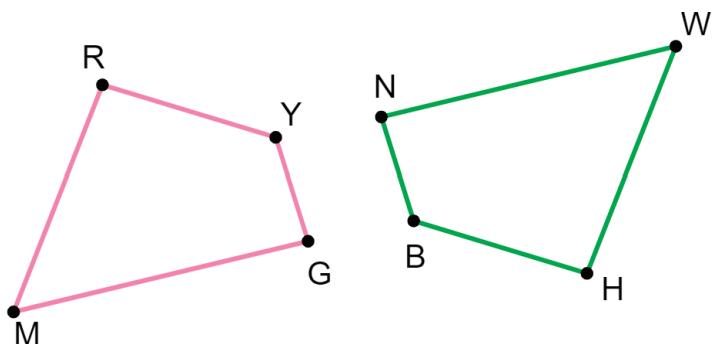
$$HW = 2x - 2, \quad XD = \frac{3}{2}x - 1, \quad HX = 2x + 5, \quad WD = 2x - 2$$

- b. Find the length of $\overline{MX}$.



5.9.5 Wrap-Up: Error Analysis

1. Quadrilaterals $RYGM$ and $HBNW$ are shown.



$RYGM \cong HBNW$ and the following information is also true.

$$m\angle RYG = (10x - 6)^\circ, MG = 2k + 3, NW = 4k - 29, \\ m\angle MGY = (7x - 5)^\circ, m\angle BNW = 86^\circ, HB = k + 9$$

Hector used the given information to write the two equations shown.

$$10x - 6 = 86$$

$$4k - 29 = 2k + 3$$

One of Hector's equations is NOT correct. Determine which one, and correct his error.

Unit 5, Lesson 10: Congruence in Two-Dimensional Figures – Part 2



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems involving congruence.



5.10.1 Warm-Up: Notice and Wonder

A table of menu prices is shown.

Burgers	Chips	Fries
Deluxe Burger	\$6.90	\$7.70
Deluxe Cheese	\$7.70	\$8.45
Bacon Burger	\$7.95	\$8.65
Bacon Cheese	\$8.25	\$8.95
Double Burger	\$8.35	\$9.30
Double Cheese	\$9.25	\$9.95
Double Bacon Cheese	\$9.50	\$10.25
Cheese Garden Burger	\$8.25	\$8.85
Mushroom Swiss	\$8.25	\$8.95
Jalapeno Pepper Jack	\$8.25	\$8.95
BBQ Western Bacon	\$9.20	\$9.85
Pastrami Burger	\$9.20	\$9.85
Power Pac	\$10.45	\$11.05
Cheese Power Pac	\$11.25	\$11.70

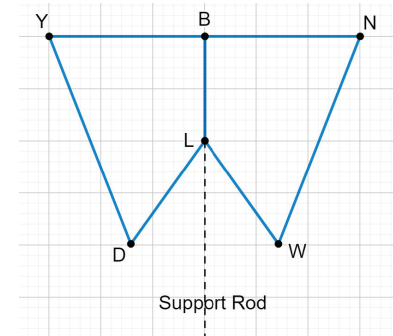
- What do you notice about prices of the burgers with cheese?
- What do you notice about ordering chips versus fries?
- There is a price difference between adding chips or fries with a double burger and adding chips or fries with cheese garden burger. Do you think this means that when you order a double burger you get more fries?
- When you look at a menu, do you consider the price differences or just order what you would like to eat?



5.10.2 Guided Instruction: Corresponding Parts in the Real World

Oscar and Aarna worked together to create a prototype obstacle for a drone obstacle course. They used grid paper to create their design as shown.

To be sure the design met regulations for the competition, they used the equations to input the design into a computer program. They discovered that some of their work disappeared, so they used the given equations knowing that $YBLD \cong NBLW$.



The perimeter of $YBLD$ is 293 centimeters (cm). $\angle YBL$ is a right angle.

$$YB = 75 \text{ cm} \quad WL = (3x + 10) \text{ cm} \quad m\angle LWN = (4k + 9)^\circ$$

$$m\angle DLB = (12k + 1)^\circ \quad WN = (6x + 5) \text{ cm} \quad LB = 50 \text{ cm}$$

$$m\angle BYD = (5k + 8)^\circ$$

- What is the value of x ?

2. What is the length of $\overline{WL}$? Include units.

3. What is the length of $\overline{YD}$? Include units.

4. What is the value of k ?

5. What is the measure of $m\angle BYD$, in degrees?

6. What is the measure of $m\angle LDY$, in degrees?

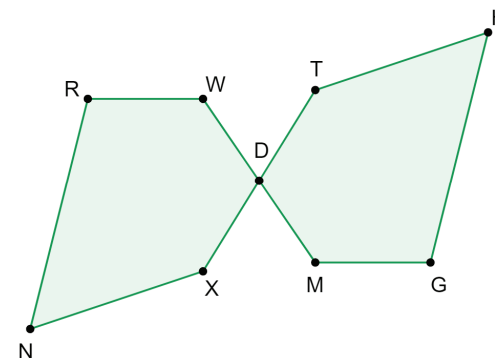


5.10.3 Collaboration: Corresponding Parts in the Real World

Work with your partner to complete the following.

French formal gardens use geometric shapes that repeat throughout the garden. Dajon is a landscape designer. He is designing a garden in the tradition of a French

formal garden. While at the client's home, he wrote down some measurements, in feet (ft), and then sketched polygons $RWDXN$ and $DTHGM$ to represent the garden. In the tradition of the congruence of polygons found in French formal gardens, Dajon also wrote $NRWDX \cong HGMDT$.



The garden's measurements are given in the table.

Garden Measurements	
$WD = (4x - 0.1)$ ft	$MD = 10.3$ ft
$RW = 12$ ft	$TH = (8x - 1.8)$ ft
$DX = (3x + 3.4)$ ft	$HG = (10x - 1.3)$ ft

1. Determine the perimeter of the garden. Show your work.

Once the garden has been installed, Dajon will check the angles of the polygons.

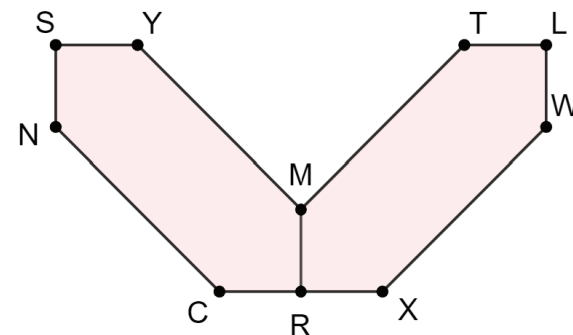
$m\angle HTD = (5k - 6)^\circ$
$m\angle XNR = 58^\circ$
$m\angle NRW = (4k + 4)^\circ$
$m\angle DMG = (6k - 20)^\circ$
$m\angle MGH = 104^\circ$

- Determine what measurement, in degrees, Dajon should use to check $m\angle XDW$. Show your work.



5.10.4 Your Turn!

- Mireia is learning how to make stained glass in art class. She has created a design and labeled the vertices to help her keep track of the measurements. Her design is shown.



Mireia's design was created so that $RMYSNC \cong RMTLWX$, $\overline{CN} \cong \overline{MY}$, and $\overline{CR} \cong \overline{RM} \cong \overline{YS} \cong \overline{SN}$. Mireia needs to determine how much solder she needs. Solder is used along the perimeter, in inches (in.), of the design to set the glass in place.

Use the given measurements, in inches, to determine the amount of solder.

$$WX = 9x - 2.3$$

$$YM = 6x + 1.3$$

$$MR = 3x - 0.6$$



5.10.5 Wrap-Up: Reflection

1. Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentence by checking all that apply.

I don't understand

- ☐ SSS
- ☐ SAS
- ☐ ASA
- ☐ AAS
- ☐ HL
- ☐ CPCTC

... YET.

- ☐ Naming congruent parts
- ☐ Paragraph proofs
- ☐ Two-column proofs
- ☐ Flowchart proofs
- ☐ Finding values in congruent polygons

do(does) NOT make sense ...
YET.

I can explain

- ☐ SSS.
- ☐ SAS.
- ☐ ASA.
- ☐ AAS.
- ☐ HL.
- ☐ CPCTC.
- ☐ naming congruent parts.
- ☐ finding values in congruent polygons.

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